Routine Computing: A Systematic Review of Sensing Daily Life Dimensions Towards Human-Centered Goals

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Abstract

Human routines structure daily life, yet remain challenging for computational systems to understand. This paper presents the first systematic review of routine computing, a previously implicit but increasingly recognized field that focuses on computationally sensing and modeling human behaviors. It synthesizes 203 studies published up to August 2025. The paper presents a new taxonomy of the literature, focusing on temporal structures, behavioral interactions, cognitive aspects, and how variability and deviations are addressed. The common goals of routine computing extend across four major application domains, including accessibility care, the promotion of healthy habits, adaptive and context-aware support, and large-scale population insights. Persistent challenges that limit the design of truly human-centered systems are identified, including the gap between low-level activity recognition and high-level intent, the tension between personalization and generalization, unresolved privacy concerns, and data-related limitations. By consolidating these findings, this paper provides a foundational framework for HCI researchers, outlining principles for designing ethical, adaptive, and human-centered routine-aware systems.

CCS Concepts

• **Human-centered computing** → *Empirical studies in HCI*; • **General and reference** → **Surveys and overviews**.

Keywords

routine computing, ubiquitous sensing, behavioral modeling, longitudinal analysis, personal informatics, human-computer interaction

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1 Introduction

Human life is structured by routines—recurring sequences of activities such as commuting, eating, working, and resting—that give rhythm, meaning, and stability to everyday experience. Recent advancements in ubiquitous computing have sought to integrate more seamlessly into these daily patterns by modeling and responding to human behavior. Yet, as Suchman's *Situated Action* [184] and Dourish's *Embodied Interaction* [49] remind us, human behavior is deeply contextual and interpretive, making the computational modeling of routines particularly demanding.

The study of human routines in computing draws on several long-standing research traditions. Human Activity Recognition (HAR) research, beginning with foundational work by Katz [99] and Lawton and Brody [115], established formal taxonomies of

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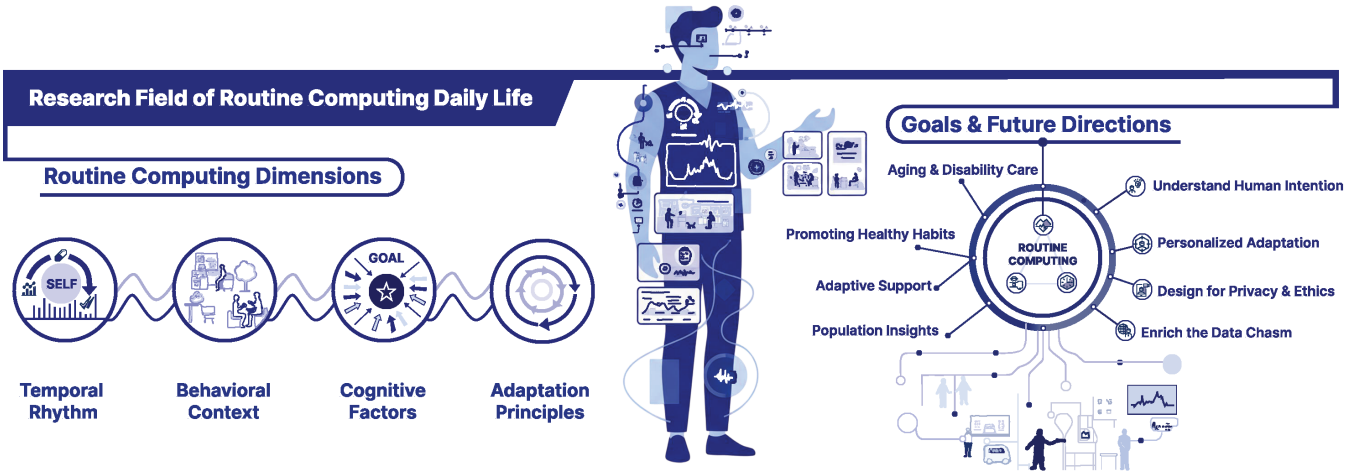


Figure 1: Visual abstract of routine computing survey and taxonomy. This framework summarizes the conceptual landscape of routine computing across three major layers. Dimensions outline four core perspectives for characterizing everyday routines: *Temporal Rhythm*, *Behavior Context*, *Cognitive Factors*, and *Adaptation Principles*. Goals describe four primary application targets, including *Aging and Disability Care*, *Promoting Healthy Habits*, *Adaptive Support*, and *Population Insights*. Future Directions highlight emerging opportunities for advancing the field through *Understanding Human Intention*, *Personalized Adaptation*, *Design for Privacy and Ethics*, and *Enriching the Data Chasm*.

self-care and instrumental tasks that later informed monitoring practices in gerontology, disability care, and home-based health assessment. Context-aware computing [1] extended this perspective by emphasizing that meaningful interaction requires sensing and interpreting the situational factors surrounding an activity—location, timing, social presence, and environmental state. Across these traditions, prior surveys have offered important overviews of related areas [13, 170], however, they were not designed to address the broader patterns that structure daily life. They typically focus on momentary or single-category activities, rather than the complex, evolving, and interdependent actions that unfold across hours, days, and weeks. To address this gap, we introduce the concept of **routine computing**, defined as the computational sensing, modeling, understanding, and application of recurring human behaviors in daily life. Routine computing extends beyond activity recognition by foregrounding the temporal, contextual, and psychological dimensions that give everyday practices their meaning and significance. Our work is guided by the following research questions:

RQ1: What conceptual dimensions are emphasized by existing studies of daily routines in HCI and ubiquitous computing research?

RQ2: What goals drive the development of routine-aware technologies?

RQ3: How do the identified dimensions and goals of routine computing shape the user-experience challenges for designing future human-centered routine-aware systems?

These research questions were formulated to clarify the foundational structure of routine computing as a research area. **RQ1** aims to identify the underlying dimensions that organize existing approaches and to establish a shared conceptual vocabulary. At the same time, routine-aware systems are developed with diverse aims—from supporting health and aging to improving productivity or understanding community behavior—yet these motivations have not been systematically compared. **RQ2** addresses this by examining the functional goals that shape the development of routine-oriented technologies and the problems they aim to solve. Finally, as routine computing increasingly influences user-facing technologies, it raises questions about how such systems should interact with people, integrate into everyday life, and support long-term engagement. **RQ3** therefore explores the design and user-experience considerations that follow from routine-aware computation. Together, these questions provide a structured basis for synthesizing the literature and defining the scope of routine computing as an emerging research domain.

Therefore, in this paper, we conduct a systematic review of routine computing in HCI and ubiquitous computing, synthesizing 203 studies up to August 2025. Our analysis classifies work by sensing temporal granularity, behavioral and social contexts, cognitive and psychological factors, and adaptation strategies. The review examines routine-based applications in domains such as aging, health, self-reflection, and population-level insights.

Accordingly, this paper makes the following contributions:

(1) We provide the first **systematic review** of routine computing in daily life, synthesizing 203 studies across sensing, modeling, and applied domains.
(2) We articulate **routine computing as a conceptual foundation**, distinguishing it from related areas such as activity recognition and context-aware computing.
(3) We develop a **taxonomy of routine computation** along temporal, behavioral, cognitive, and adaptive dimensions, consolidating diverse methods and approaches.

(4) We identify **critical challenges and future opportunities**, including interpretability, personalization, ethical design, and data enrichment, setting a research agenda for advancing routine-aware HCI.

2 Methodology

2.1 Coding

This review follows PRISMA 2020 guidelines [153], ensuring a rigorous and transparent process for identifying and analyzing studies on routine computing in daily life. In accordance with the guidelines, this section consists of four key steps: 1) independent article screening by two researchers, 2) pilot annotation to identify key information for analysis, 3) iterative coding and establishment of a systematic codebook, and 4) synthesis of results and finalization of findings.

2.2 Research Strategy

Initially, we conducted an exploratory survey of the routine-computing research landscape to identify the terminology most consistently used across relevant domains. We examined a total of 30 search results generated by different combinations of routine-related terms (e.g., "routine(s)", "daily life", and "ego-centric") together with behavioral and computational descriptors (e.g., "pattern", "behavior", and "modeling"), as well as domain-specific terms related to ubiquitous and context-aware computing. This iterative process ensured that our final keyword set aligned with how routine-oriented work is described across health, smart-home, mobility, and lifelogging research. For the final selection, we prioritized terminology appearing in foundational and frequently cited papers on ubiquitous computing, activity recognition, and emerging context-driven systems. Furthermore, the selection was supplemented by our established definition of the term "routine computing". Each group was combined with an "AND," yielding a total of 2,631 records (574 from ACM DL and 2,057 from IEEE Xplore). Next, a screening process was conducted to select papers within the survey's scope.

We restricted our search to the ACM Digital Library and IEEE Xplore, as our review focuses on practical, technology-oriented routine computation: systems that involve sensing, modeling, prediction, and empirical evaluation. These technical contributions are overwhelmingly published in HCI, ubiquitous computing, and pervasive sensing venues, which are concentrated within ACM and IEEE. Preliminary searches in SpringerLink, ScienceDirect, and PubMed/NIH showed that these databases primarily include biomedical, clinical, or theoretical work that does not involve computational routine modeling or deployable systems. We did not specify a time interval for the search, and the last search was conducted in August 2025.

2.3 Screening

We evaluated the relevance of all 2,631 papers returned by our search queries based on their titles and abstracts. Two authors independently screened all references in parallel and applied the following criteria for inclusion and exclusion: (1) we included only papers written in English; (2) we included only full papers published in peer-reviewed journals, conferences, symposiums, or workshops, thereby excluding technical reports, dissertations, work-in-progress papers, posters, patents, and tutorial or workshop proposals; (3) we filtered for topical match following our definition of routine computing and included papers that analyzed human behavior longitudinally, modeled recurring activity patterns, or used sensing technologies to characterize daily routines; (4) we excluded papers in which the term "routine" referred to technological or system-level processes such as network routines, workflow automation, cybersecurity routines, robotic task loops, or database maintenance; and (5) we excluded papers involving non-human subjects, including robotic, industrial, or animal behavior studies.

Table 1: Final search keywords used in the literature review (combined with AND).

Keywords (combined with AND)
Routine-related terms: "routine(s)", "daily life", "daily routine", "ego-centric", "proactive"
Behavioral descriptors: "activities", "pattern", "behavior", "user model(s)"
Computational approaches: "discover", "modeling", "recognition", "understand", "lifelog", "monitoring", "activity tracking"

2.4 Eligibility

After title and abstract screening, 396 papers remained for full-text assessment. To determine eligibility for full inclusion after title and abstract screening, the same two authors independently reviewed all candidate papers at the full-text level, using the following additional criteria: (1) we included only papers providing clear methodological detail on sensing, modeling, or evaluating recurring human behaviors, excluding works that referenced routines without computational treatment; (2) we included only primary research contributions and excluded surveys, theoretical essays, opinion pieces, and purely conceptual frameworks; (3) we excluded papers that did not address recurring or temporally structured daily-life behavior, such as studies focused solely on algorithm optimization tasks, short-duration activities in a pre-set laboratory environment, or one-shot predictions without evaluation; (4) we excluded studies whose contributions were redundant with earlier work, for example follow-up papers reusing identical datasets and methods without offering substantial new insights.

Following this eligibility assessment, 203 papers (77 from ACM DL and 126 from IEEE Xplore) were retained for analysis. Inter-rater reliability between the two reviewers was evaluated using Cohen's κ, which indicated substantial agreement ($\kappa = .833$, 95% CI [.779, .888], $p < .001$). Any remaining disagreements between reviewers were discussed and resolved before final inclusion.

To systematically extract information from the corpus, we developed a structured data-extraction sheet consisting of 36 columns. The sheet was created through an initial open-coding phase in which both authors independently reviewed a subset of papers to identify recurring methodological and conceptual features relevant

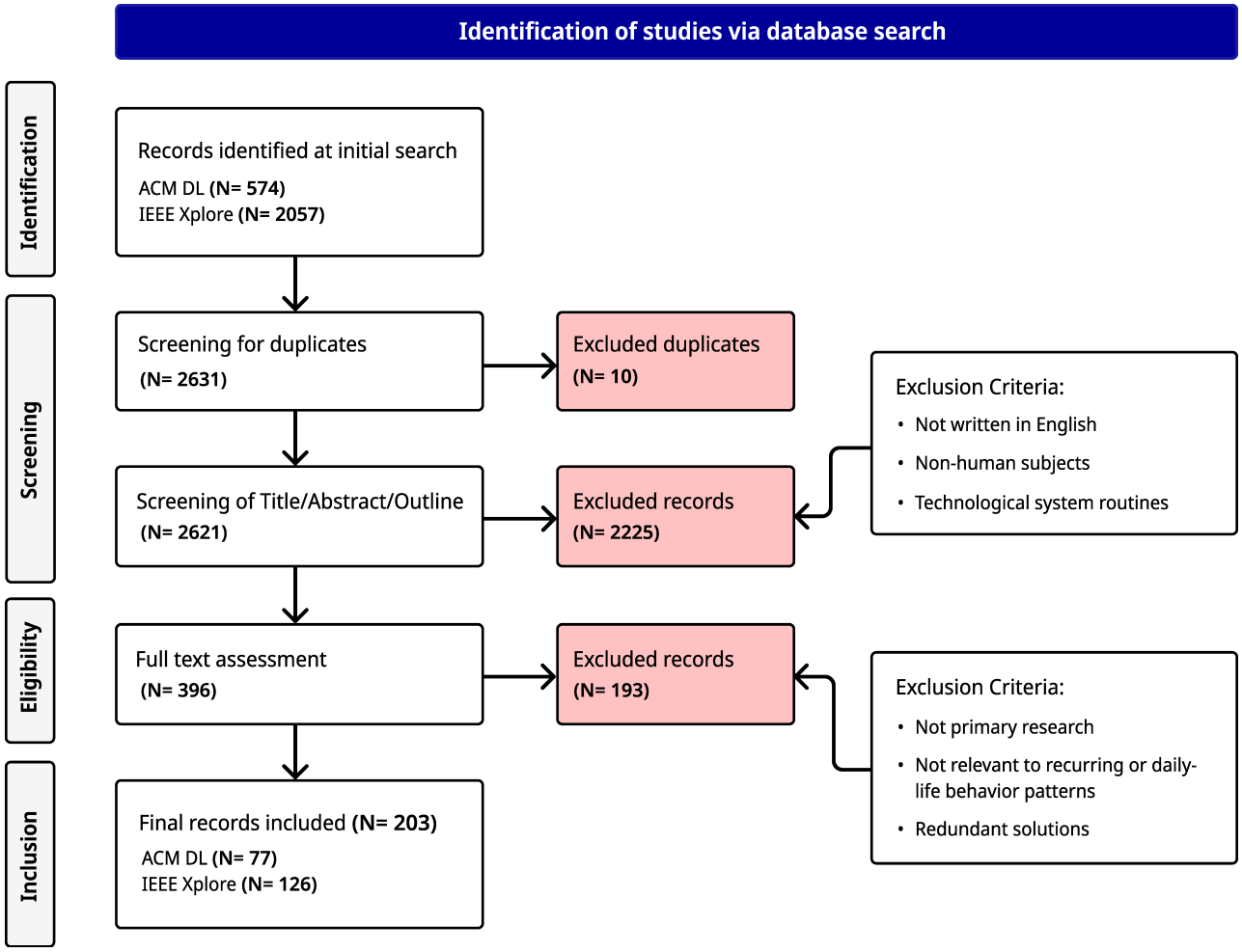


Figure 2: PRISMA flowchart detailing the article selection procedure

to routine computing. Codes were iteratively discussed and refined until consensus was reached. The extraction schema captured several key aspects: (1) *basic metadata* to contextualize each paper's scope and contribution; (2) *sensing strategies and data-collection methods* to examine assumptions about how routines are captured; (3) *modeled activities and computational techniques* to understand how routines are operationalized; (4) *user-study characteristics and cognitive considerations* to reveal how human experiences and interpretations are incorporated; (5) *evaluation metrics* used to assess performance, usability, or model quality; (6) *the problem domain* to link routine-focused system goals to real-world applications; and (7) *reported challenges* to identify recurring limitations across the field. This structure ensured that the coding remained aligned with the research questions: **RQ1** focused on how routines are conceptualized and represented, while **RQ2** and **RQ3** guided attention to human-centered adaptation, usability, and evaluation practices. Importantly, by allowing patterns to emerge inductively from the corpus, this process enabled the identification of potential conceptual dimensions described in the next section.

3 Dimensions of Human Routine Computation[1]

This section provides an integrated overview of human routine computation across 203 studies, structured around a set of core dimensions that emerged consistently throughout the corpus. The derivation of these dimensions followed an inductive–comparative process grounded in the coding framework developed during study extraction. Although the reviewed works vary in populations, sensing modalities, and analytical objectives, their approaches recurrently converged on several foundational features: the temporal organization of behavior, the types of activities and interactions being recognized, the cognitive or motivational processes underlying those behaviors, and the mechanisms through which systems adapt to change over time.

Across comparative analysis, these recurring features aligned robustly with four major analytical focuses that collectively reflect the conceptual pipeline of routine computing:

[1]In this section, *n* refers to the total number of category instances identified across all reviewed papers, rather than the number of papers. (e.g., one paper can contribute more than one method category).

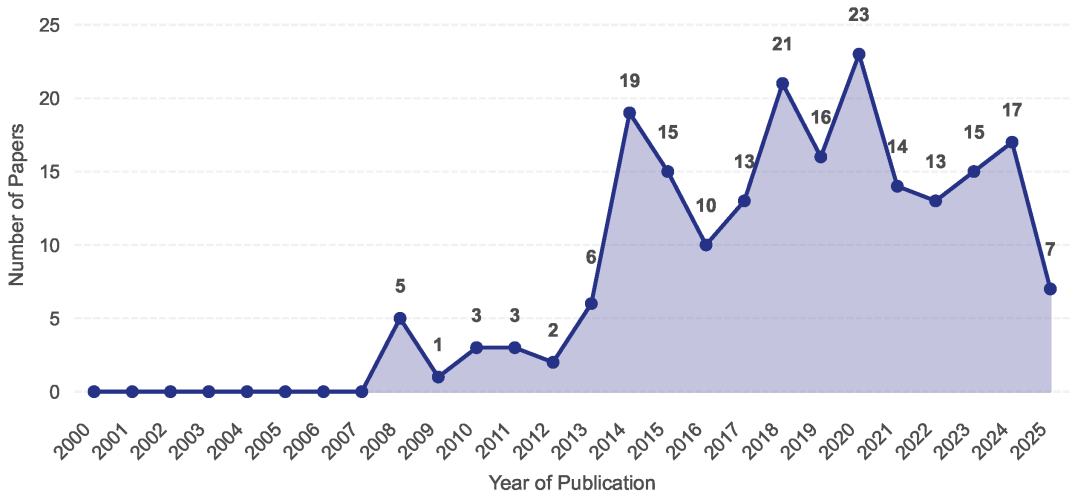


Figure 3: Annual distribution of included routine-computing publications that fulfill our criteria, illustrating the field's expansion since 2008. The chart only includes papers published up to August 31, 2025.

- **Temporal granularity**—capturing when routines occur and how frequently they repeat, thereby revealing the rhythmic structure of everyday behavior.
- **Behavioral interaction context**—characterizing common activities and interactions with objects, bodies, and the surrounding environment.
- **Cognitive and motivational factors**—explaining the effect of integrating goals, intentions, and cognitive effort into routines.
- **Variability, adaptation, and breakdown detection**—describing how routines evolve, deviate, or reorganize in response to internal or external change.

The following sections examine each dimension in depth, outlining how it reflects a core dynamic of daily behavior—from temporal rhythms to contextual interactions.

3.1 Temporal Granularity of Sensing Data (n=203)

Capturing routines always requires attention to time, as the meaning of behaviors depends on their time organization and sequencing. We identified three levels of temporal granularity: longitudinal lifestyle analysis, multi-day routine analysis, and moment-to-moment actions, each reflecting different research goals, sensing strategies, and application contexts. A time distribution overview of the reviewed studies is presented in Table 2.

Table 2: Distribution of reviewed papers by collection time (six granular bins).

Collection time	Papers	Total (n)
Seconds-Minutes	[7, 10, 14, 16, 21–23, 27, 35, 36, 42, 43, 46, 51, 52, 54, 55, 57, 59, 60, 78, 79, 82, 84, 90, 91, 93, 97, 109, 116, 120, 127, 134, 140, 143, 144, 149, 151, 165, 167, 188, 190, 191, 194, 198, 199, 205, 208, 211, 213, 218, 221, 225, 226]	54
Hour - Day	[37, 77, 88, 92, 96, 113, 129, 160, 169, 192, 222]	11
Days - Week	[8, 20, 24, 26, 29, 80, 83, 87, 94, 103, 105, 106, 112, 114, 146, 175, 178, 202, 212]	19
Weeks - Month	[4, 9, 11, 15, 18, 32, 38, 48, 62, 66, 68, 70, 71, 76, 95, 101, 108, 111, 117, 123, 131, 133, 135–137, 139, 156, 159, 164, 172, 173, 185, 189, 197, 201, 203, 209, 215, 223]	39
Months - Year	[2, 3, 5, 6, 19, 28, 30, 31, 33, 34, 39, 45, 47, 50, 53, 56, 61, 63–65, 67, 74, 75, 81, 89, 98, 100, 102, 104, 107, 110, 118, 119, 122, 124–126, 128, 130, 138, 141, 142, 148, 154, 155, 158, 161–163, 166, 174, 176, 179, 181, 183, 186, 193, 196, 210, 214, 216, 217, 220, 224]	65
More than 1 year	[12, 25, 72, 73, 86, 121, 147, 150, 171, 177, 195, 200, 206, 207, 219]	15
	Corpus total	**203**

3.1.1 Longitudinal Lifestyle Analysis (n=80). Longitudinal studies reveal how routines develop over months or years, showing how repeated behaviors form lifestyle patterns. Low-burden, large-scale sensing is particularly effective for detecting gradual changes in health, energy use, or mobility that may be missed in daily observation. A group of studies ($n = 18$) focuses on long-term monitoring of sleep, stress, and mental states through continuous physiological monitoring [6, 126, 150, 207]. Another ($n = 19$) tracks household appliance usage and domestic routines via high-frequency sensing, including smart meters for electricity and gas, with environmental sensors capturing temperature, humidity, lighting, CO_2, sound, and pressure to contextualize activities such as showering [30, 47, 104, 147, 166, 174]. Mobility routines ($n = 17$) are monitored indoors and outdoors through GPS, cellular, BLE beacons, Wi-Fi CSI, PIR, and LiDAR, capturing movement patterns without labeling specific activities [12, 17, 19, 119, 121, 158, 161]. Furthermore, 30% of the studies ($n = 26$) analyze individual and population behaviors from large-scale digital traces such as social media, campus card

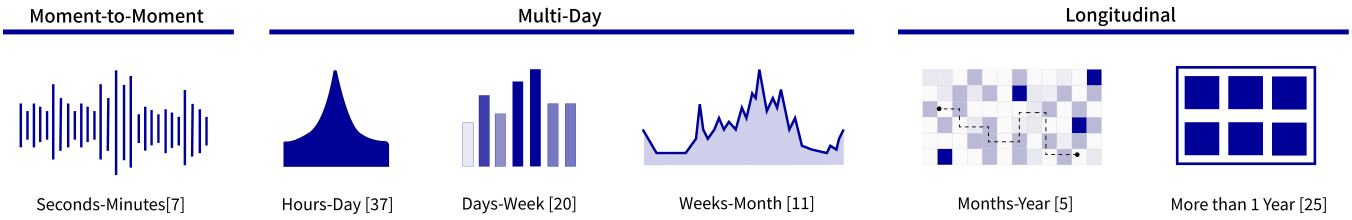


Figure 4: Instances demonstrating different temporal granularity patterns in sensing data.

use, location-driven patterns, and call logs to reveal urban mobility, consumption, and social routines [56, 100, 110, 124, 128, 217].

3.1.2 Multi-day Routine Analysis (n=69). To capture daily rhythms over days or weeks, this temporal perspective reveals how routines stabilize, adapt, or break down, shedding light on habit formation, daily structure, and rhythm deviations. Studies on personality and behavioral traits ($n = 17$) analyze multi-day patterns of device use, physical mobility, and social interaction to model stable tendencies with individual adaptation [18, 62, 71, 101, 103]. Other work examines repeated routine action units such as meals, work, leisure, household tasks, exercises, and hygiene to uncover habitual structures and daily rhythms [11, 37, 66, 76, 77, 87, 94, 169, 202]. Cognitive and affective daily patterns ($n = 16$) capture fluctuations in mood, stress, and attention through activity and behavioral data. They can detect sudden surges or spikes in emotional states, highlighting moments when routines are disrupted and enabling timely interventions in situations that require immediate attention or control [4, 8, 114, 131, 135, 209]. Research on habit formation and adjustment monitors recurring behaviors to identify emerging patterns and stability [15, 38, 173]. Personalized self-reports ($n = 2$) enhance people's ability to remember time-sensitive or health-critical activities [24, 105]. Finally, a major portion ($n = 34$) focuses on health and activity monitoring of elderly or vulnerable populations, such as visually impaired people, capturing temporal patterns in daily structure, physiological rhythms, and interactions with objects or appliances [160, 197, 203, 223].

3.1.3 Moment-to-Moment Actions (n=54). Momentary analyses focus on micro-scale human behaviors, where gestures, postures, and rapid adjustments unfolding in seconds or minutes reveal fine-grained patterns of routine. Hand and arm movements ($n = 14$) are tracked via wearable IMUs, smart rings, and gloves to capture gestures, grip adjustments, and object manipulations, enabling assessment of motor skills and task precision [21, 57, 211, 221]. Whole-body micro-movements ($n = 9$) are monitored using IMUs, pressure mats, and multi-sensor wearables, recording balance corrections, stance shifts, and micro-gait variations to evaluate stability and fatigue [36, 51, 54, 167, 188, 226]. Short, repetitive, or rhythmic actions ($n = 9$), such as typing, tapping, or exercise repetitions, are captured with high-frequency accelerometers and wrist-worn devices to assess temporal consistency and performance [91, 120, 190]. Rapid locomotion and contextual micro-patterns ($n = 22$) examine moment-to-moment positional adjustments: some studies track short bursts of stepping, sidestepping, or gait corrections via IMUs and GPS [52, 90, 205, 225], while others capture quick directional changes using environmental sensors, infrared sensors, or microphones with continuous event detection [23, 59, 109, 140]. Video-based surveillance focuses on abrupt body movements to identify dangerous shifts that precede falls [27, 60, 149].

3.2 Behavioral Interaction Contexts (n=170)

Routines are sustained through recurring behavioral interactions that span multiple levels of daily life, including immersion in the self, the body, the surrounding environment, and other people. These interactions unfold through habitual practices of personal health and wellness[145]. They are also expressed in embodied movements, shaped by the material and spatial advantages of technology and the surrounding environment [69]. At the same time, they emerge as socially situated practices that derive meaning and stability from collective participation and cultural context [180]. Together, these aspects highlight how routines are more than mere repetition but dynamic patterns that intertwine personal, material, and social layers of experience. Table 3 summarizes the core aspects identified in the reviewed studies, illustrating how researchers have operationalized and examined different behavioral layers of routine engagement.

3.2.1 Self-Care Interactions (n=66). Research on habits indicates that self-care routines often run on "autopilot", with hygiene, nourishment, and sleep reducing cognitive effort for complex tasks [145]. Focused personal hygiene ($n = 15$) includes showering, handwashing, and brushing teeth, supporting health and social readiness [15, 29, 82, 171, 223]. Incidental self-care ($n = 6$) covers need-driven actions such as toileting and hydration, where frequent and unpredictable interruptions shape daily patterns [28, 76, 142, 177]. Nutrition management ($n = 7$) involves food preparation and cleanup, where deviations reveal cognitive, social, or motivational changes [5, 62, 131, 148, 218], while nourishment consumption ($n = 2$) captures eating and drinking patterns that structure routines and foster social participation [7, 200]. Restorative sleep ($n = 16$) reflects the quality of habitual rest, which strongly influences physical and cognitive function [32, 53, 195, 207]. Environmental maintenance ($n = 13$), such as cleaning, vacuuming, and organizing of living space, signals physical or motivational capacity [94, 133, 146, 176]. Medication management ($n = 7$) emphasizes the adherence to rule-based and contextual reasoning based on observation of events, where timing is critical for health outcomes [27, 31, 89, 91, 97, 135].

3.2.2 Physical & Mobility Interactions (n=71). From an ecological perspective, mobility is more than movement through space. It reflects how people act on environmental opportunities, revealing

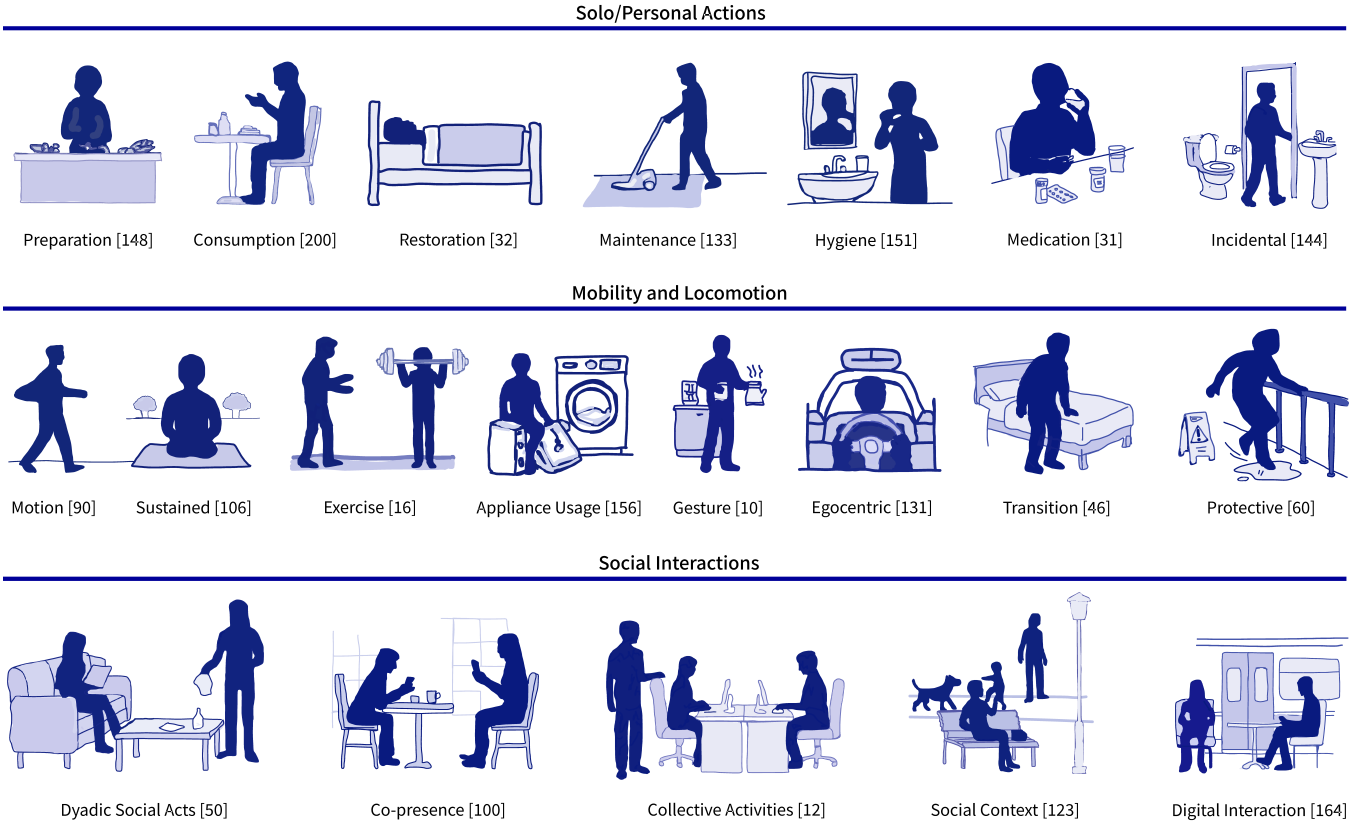


Figure 5: Instances demonstrating different behavioral interaction contexts.

both capabilities and constraints [69]. Ambulation and locomotion ($n = 11$) capture goal-directed movements such as walking, running, or climbing stairs, where path and gait indicate ability and fatigue [18, 51, 84, 90, 96]. Postural transitions ($n = 5$), such as getting out of bed, highlight how speed and stability shape the initiation of activities [113, 178, 198, 213]. Sustained body positions ($n = 6$) reflect comfort, engagement, or health risks by location and duration of posture [55, 79, 106, 143, 188]. Exercise and deliberate movement ($n = 6$) cover structured activities such as workouts, yoga, or rehabilitation, where consistency and form reveal motivation and intensity, indicating whether individuals promote their health [16, 43, 48, 154, 169, 189]. Object manipulation ($n = 27$) involves handling items—pouring, stirring, writing—that combine into larger tasks like cooking or drinking [83, 88, 203]. Appliance and device-centered interactions ($n = 7$) include daily operations of lights, stoves, or kettles [30, 47, 156]. Anomalous and protective movements ($n = 5$), such as falls or bracing, signal immediate safety risks [36, 60, 68, 199, 221]. Finally, egocentric recording ($n = 4$) captures fine-grained body movements and attention shifts that are often missed by stationary or device-centric sensors [24, 25, 60, 131].

3.2.3 Social Interactions (n=33). Social practices are fundamental to routines, as daily life is organized not only by individual habits but also by shared norms, roles, and collective activities [157, 168]. Dyadic social acts capture one-to-one encounters, such as handshakes, greetings, and phone calls ($n = 6$), reflecting direct physical or verbal exchanges between two people [24, 50, 107, 155]. Group co-presence and co-location ($n = 5$) emphasize shared spatial proximity, such as class or workplace attendance inferred from Bluetooth or mobility data [56, 100, 186, 197, 224]. Collective social activities ($n = 7$) involve coordinated participation in leisure or community events, including shopping, parties, concerts, dining, and therapy sessions, or opportunistic encounters based on shared interests [124, 149, 158, 161]. Social context within routines ($n = 5$) highlights how the presence of partners, family, or colleagues shapes choices of daily activity and role-based behaviors [110, 123, 215]. Mediated or digital social interaction ($n = 10$) involves technology-based exchanges such as messaging, planning, and documentation mediated by devices [101, 175].

3.3 Cognitive and Psychological Factors (n=83)

Routines are shaped by cognitive and psychological processes that influence how people remember, sustain, and reflect on their actions. We identify five factors: prospective memory, intention and motivation, reward sensitivity, cognitive effort, and self-reflection,

Table 3: Summary of behavioral interaction subcategories in routine computing.

Main Category	Sub-Category	Paper IDs (Count)
Solo / Personal Actions	Focused personal hygiene	[11, 15, 26, 29, 33, 82, 92, 95, 102, 126, 144, 151, 171, 183, 223] (15)
	Incidental and mandatory self-care	[28, 70, 76, 98, 142, 177] (6)
	Nourishment preparation (IADL)	[4, 5, 62, 131, 148, 194, 218] (7)
	Nourishment consumption	[7, 200] (2)
	Restorative sleep	[8, 32, 34, 53, 65, 80, 81, 86, 141, 181, 195, 196, 207, 208, 210, 219] (16)
	Environmental maintenance (IADL)	[2, 20, 27, 38, 71, 94, 133, 134, 146, 147, 159, 165, 176] (13)
	Medication management (IADL)	[31, 89, 91, 97, 108, 135, 193] (7)
Mobility and Locomotion	Ambulation and locomotion	[14, 18, 22, 51, 52, 84, 90, 93, 96, 118, 167] (11)
	Postural transitions	[46, 113, 178, 191, 198, 213] (5)
	Sustained body positions	[55, 79, 106, 143, 188] (6)
	Exercise and deliberate movement	[16, 43, 48, 154, 169, 189] (6)
	Gestures and object interaction	[9, 10, 35, 39, 42, 57, 83, 88, 112, 117, 119, 121, 127–129, 138–140, 162, 179, 203, 205, 206, 211, 214, 225] (27)
	Appliance and device-centric	[30, 47, 67, 104, 125, 156, 172] (7)
	Anomalous and protective movements	[36, 60, 68, 199, 221] (5)
	Egocentric motion view	[24, 25, 60, 131] (4)
Social Interactions	Dyadic social acts	[24, 25, 50, 107, 120, 155] (6)
	Group co-presence and co-location	[56, 100, 186, 197, 224] (5)
	Collective social activities	[12, 124, 149, 158, 161, 212, 220] (7)
	Social context within routines	[72, 73, 110, 123, 215] (5)
	Mediated or digital social interaction	[23, 101, 136, 137, 163, 164, 174, 175, 192, 216] (10)

which together explain how routines connect to memory, meaning, and personal values. An overview of these factors is summarized in Table 4.

3.3.1 Remembering Future Tasks (Prospective Memory) (n=4). Prospective memory theory emphasizes how systems support users in keeping future intentions in mind, such as taking medication or attending scheduled meetings[132]. For instance, one study provides reminders for deviations from expected routines [155], while another predicts users' upcoming actions in real time to anticipate intended behaviors [5, 185]. Moreover, prompting users to complete self-reports enhances accurate recall of daily activities [105]. These approaches highlight the importance of external aids in helping users remember what they plan to do, rather than relying solely on immediate cues.

3.3.2 Intent and Intrinsic Motivation (n=4). Self-determination theory emphasizes that routines sustained by intrinsic motivation linked to autonomy, purpose, or enjoyment are more resilient than those driven solely by external demands, highlighting the need for systems that align with personal goals and values[44]. Systems that generate personalized "if-then" plans link situational cues to desired behaviors, strengthening intention-driven actions [48]. Others examine intrinsic motivations for eating [66] or exercising [154], revealing how pleasure, health, and personal goals drive adherence. Research also highlights how feelings of competence or incompetence influence participation, showing the psychological difference between a task users feel compelled to do versus one they truly want to pursue [71].

3.3.3 Motivation and Reward Sensitivity (n=9). Research in behavioral psychology[182] shows that routines are strengthened when actions are paired with rewarding outcomes, whether through pleasure, achievement, or relief, underscoring how technologies can harness reinforcement to maintain engagement. Measurement of emotional states, stress, and arousal [26, 114, 183, 186] exemplifies how routines align with positive or negative outcomes as well as the duration and intensity of the experience. Research [101, 124, 217] highlights how social influences and individual personality traits affect the satisfaction and achievement of daily activities. Others leverage reward sensitivity to support behavior change, such as smoking cessation [7] or interventions for mental well-being [150], showing how systems can strategically harness motivational levers to maintain engagement.

3.3.4 Mental Effort in Routines (Working Memory and Cognitive Load) (n=26). Cognitive load theory suggests that routines falter when tasks demand too much memory or attention, positioning automation and context inference as crucial ways to ease effort and support sustainable behaviors [187]. Systems that automate activity recognition [22, 129, 171], segment complex action sequences [35, 43, 64], or infer hierarchical context [61, 122, 134, 144] reduce cognitive load, allowing users to execute routines without being overwhelmed. Some focus on eye-context analysis to reveal which routine steps strain visual attention.[24] Others simplify repetitive or attention-demanding tasks, such as automatically adjusting smartphone settings [83] or detecting object interactions [202], freeing working memory to support smoother and more manageable routines.

3.3.5 Reflecting on One's Own Routines (Metacognition and Self-Monitoring) (n=40). Feedback from self-monitoring tools that illustrate vital patterns enables adaptive behavior and often changes people's routines, as metacognitive research [58] shows. Approximately 63% of the cognitive studies highlight how systems enable users to notice and rethink their habits. Some focus on providing a feedback loop for self-reflection [108, 138, 139, 141, 210]. Others collect and model personal data, such as movement, smartphone usage, or digital footprints, to provide users with insights about their

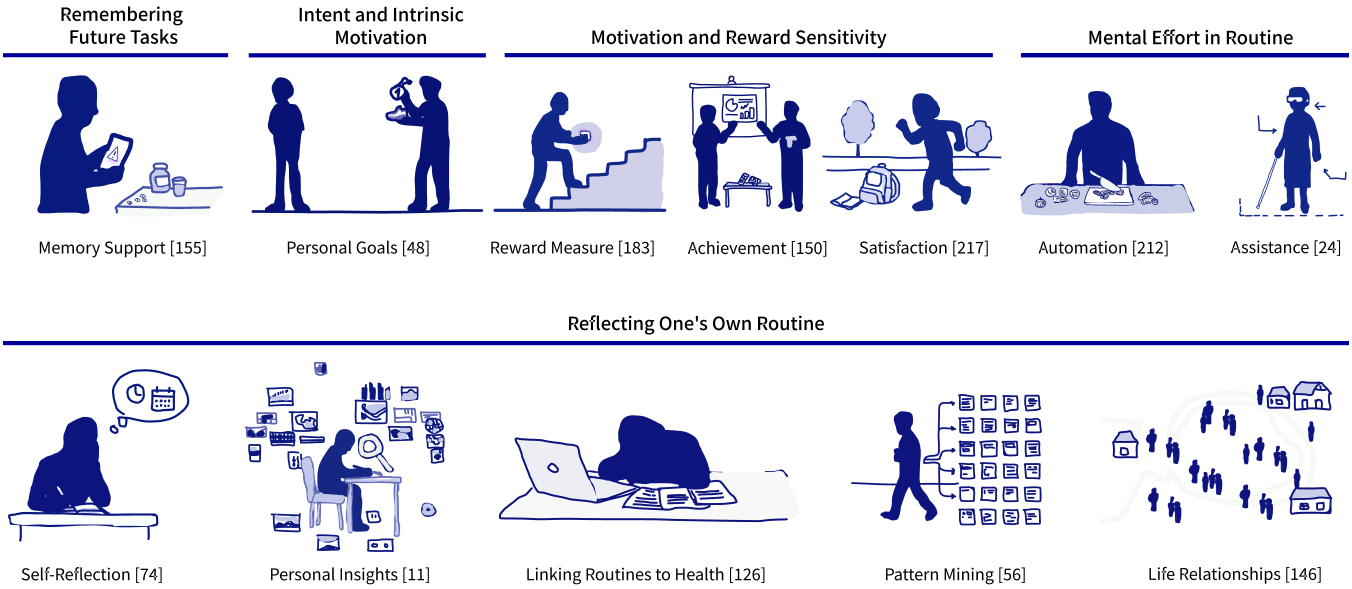


Figure 6: Instances demonstrating different cognitive and psychological factors.

Table 4: Summary of cognitive and psychological subcategories in routine computing.

Main Category	Sub-Category	Paper IDs (Count)
Remembering Future Tasks	Prospective memory support	[5, 105, 155, 185] (4)
Intent and Intrinsic Motivation	Linking actions to personal goals	[48, 66, 71, 154] (4)
Motivation and Reward Sensitivity	Measuring affective rewards	[26, 114, 183, 186] (4)
	Leveraging reward sensitivity for behavior change	[7, 150] (2)
	Satisfaction, achievement, and pleasure in routines	[101, 124, 217] (3)
Mental Effort in Routines	Automating complex activities	[12, 20, 22, 33, 43, 47, 51, 64, 72, 80, 95, 106, 107, 129, 171, 212, 223] (17)
	Reducing cognitive load through context inference	[24, 35, 61, 83, 102, 122, 134, 144, 202] (9)
Reflecting on One's Own Routines	Self-reflection feedback loop	[30, 70, 74, 108, 121, 138, 139, 141, 147, 156, 177, 207, 210] (13)
	Modeling personal data for self-insight	[8, 11, 23, 116, 123, 136, 149, 164, 196, 197] (10)
	Pattern mining and routine discovery	[56, 100, 174, 175, 224] (5)
	Linking routines to health and cognitive states	[98, 126, 135, 160, 195, 208] (6)
	Quantifying life rhythms and causal relationships	[17, 18, 32, 146, 148, 215] (6)

habits [8, 23, 149, 197]. Systems that mine patterns [174, 175, 224] or expose location-dependent bottlenecks[56, 100] in daily tasks help users pinpoint exactly where their routines break down and why. Linking routines to health outcomes (e.g., stress, cognitive decline, sleep quality, and aerobic activity) allows users to understand the impact of manner on well-being [98, 126, 160, 171, 195]. Advanced approaches quantify life rhythms [32, 148] or model causal relationships between context and action [17], supporting deeper metacognitive awareness and enabling informed behavior change.

3.4 Variability, Adaptation, and Breakdown Detection in Routines (n=83)

This dimension considers the factors driving routine deviations, along with intervention strategies for breakdowns and models of adaptation over time. An overview of the reviewed studies is summarized in Table 5.

3.4.1 Core variables of routine deviation criteria. The highlighted variables that structure routines are the specific temporal, contextual, and behavioral dimensions. Common distinctions include weekday vs. weekend schedules, activity ordering, and contextual modifiers such as weather, area of interest, or social presence

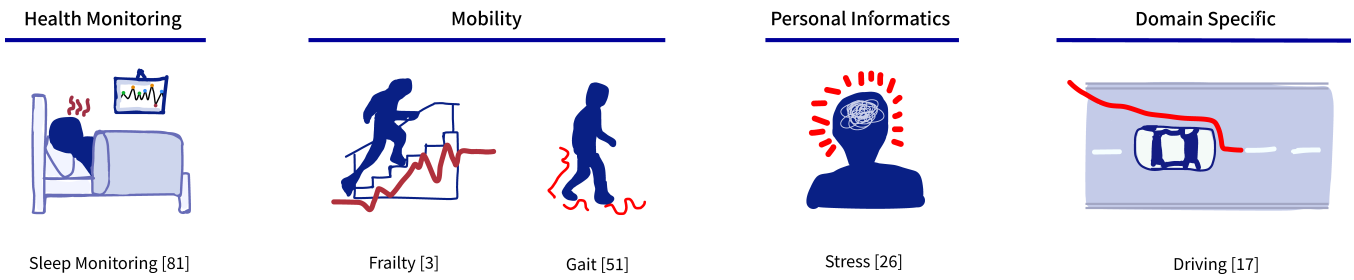


Figure 7: Instances demonstrating different variability, adaptation, and breakdown detection in routines.

[17, 94, 139, 219, 220]. Others emphasize performance-based indicators such as frailty, stress levels, or sleep patterns as critical variables for understanding deviations [26, 81, 141, 195]. Such factors are often incorporated into models to capture the fluidity of routines and inform adaptive systems. Some works emphasize domain-specific criteria, such as falls detected through accelerometer thresholds or abnormal nutritional expenditure in student routines [51, 199, 200].

3.4.2 Intervention strategies for routine support. During the anomaly detection process, the systems are set to act accordingly in case of an emergency. Many provide alerts or notifications to caregivers, family, or clinicians [3, 193, 207, 219]. Others intervene directly through reminders, feedback, or personalized recommendations to nudge user behavior [11, 48, 150, 195]. A smaller group explores novel interaction strategies, such as assistive robots for confirmation feedback [210] or visualization tools to reframe disruptive behavior [18].

3.4.3 Principles for long-term model adaptation. Addresses how systems evolve alongside changing routines. Techniques include dynamic clustering and feedback loops to incorporate new data [5, 139, 210], periodic pattern mining that adapts to contextual shifts, such as seasons or semesters [12, 89], and semi-supervised retraining approaches to handle gradual change [108, 146]. Using sliding-window multi-HMMs, the model adapts to variable routines, helping people with chronic illnesses to track medication and self-care tasks even on highly irregular days [6, 100, 166]. Other models explicitly incorporate user input or contextual queries, enabling digital twins to adapt flexibly to changes [107, 121].

3.5 Conceptual Framework for Routine Computing

Figure 8 illustrates the three core dimensions—*Context*, *Cognition*, and *Adaptation*—bounded by the outer layer of *Temporal Rhythm*, which forms the conceptual foundation of routine computing.

The outer, encapsulating role of *Temporal Rhythm (3.1)* is intentional because time and temporal structure are the practical foundation of routine computing: every system in our corpus depends on sensing streams (temporal traces of behavior, location, physiology, and device logs) to reveal repetition, periodicity, and change. Framing temporal rhythm as the prerequisite layer highlights sensing as the data substrate of routine intelligence and explains why temporal granularity determines the feasibility of analyses and interventions [1, 204].

The linkage between *Context (3.2)* and *Cognition (3.3)* is shown as interdependent because cognitive models of routine behavior do not operate in a vacuum—interpretation, intention, and memory are shaped by the environment and the available contextual cues. Accurate mental models (prospective memory, goals, workload)

Table 5: Categorization of anomaly detection papers by primary application domain.

Application Domain	Paper IDs and Key Characteristics
Health Monitoring	**Papers:** [3, 5, 6, 27–31, 60, 80, 81, 86, 94, 98, 105, 107, 121, 135, 139, 141, 155, 158, 160, 169, 171, 177, 190, 193, 203, 207, 210, 214, 219] (33 papers) **Key Variables:** Room transitions, appliance usage, sleep patterns, medication intake, and frailty indexes. **Anomalies:** Missed activities, unusual inactivity, deviations from long-term routines.
Mobility	**Papers:** [2, 11, 12, 36–38, 47, 48, 51, 56, 87, 89, 104, 108, 143, 146, 156, 160, 169, 185, 198, 199, 216, 220, 221] (25 papers) **Key Variables:** Acceleration, posture, location, pressure, WiFi CSI signals. **Anomalies:** Falls, unstable gait, abnormal transitions (e.g., sit-to-stand).
Personal Informatics	**Papers:** [17, 24, 26, 32, 37, 56, 59, 62, 71, 77, 100, 103, 114, 117, 131, 136, 149, 150, 154, 195, 200, 217] (22 papers) **Key Variables:** Sleep stages, physiological signals (HR, EDA), meal patterns, exercise, screen time. **Anomalies:** Poor sleep quality, high stress, unhealthy eating, sedentary behavior, and goal failure.
Domain-Specific	**Papers:** [57] (Handwriting Analysis), [116] (Youth Behavior), [18] (Driving Behavior) (3 papers) **Key Variables:** Pen gestures, step counts, driving maneuvers. **Anomalies:** Tremors, unusual student activity, aggressive driving.

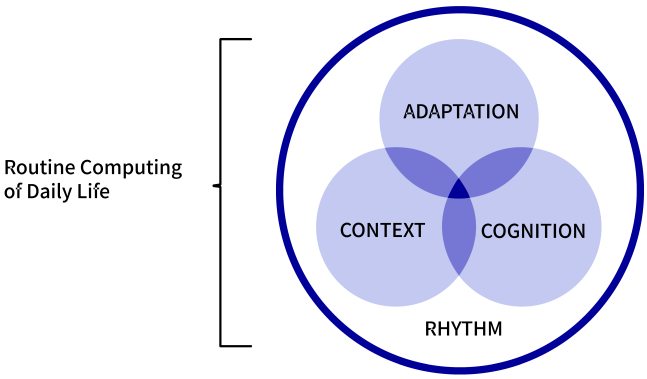


Figure 8: Conceptual framework diagram showing interdependent Context, Cognition, and Adaptation dimensions within the overarching layer of Temporal Rhythm, which enables routine computation.

require context to ground hypotheses about what an actor can or will do. This idea is supported by work on embodied and distributed cognition and by classical theories of affordances [40, 85]. Showing cognition nested with and tied to context thus makes explicit that sensing, situational cues, and mental modeling are a single epistemic chain.

Finally, *Adaptation (3.4)* is drawn as bidirectional and permeable because adaptive interventions both depend on—and actively alter—contextual inputs and cognitive states. Changes produced by adaptive systems (reminders, rescheduling, nudges, personalized feedback) can modify how often behaviors occur, which contextual cues are salient, and how users internally represent tasks or goals; conversely, shifts in context or cognition require new adaptation [145, 152]. Representing adaptation as a cross-cutting, change-producing mechanism highlights its dual role as an analytic target and as an instrument for influencing the other dimensions, aligning with empirical and theoretical work on habit dynamics and behavior-change design. Innovation often arises at the intersections of these dimensions, enabling researchers to define novel design questions that go beyond single-focus solutions. For instance, analyzing the *Context* ∩ *Cognition* relationship can inspire models that use rich sensing to predict human intention within a routine, while the *Cognition* ∩ *Adaptation* link can guide the design of interventions whose intrusiveness is dynamically tuned by the system's prediction certainty.

By making these hierarchical and functional relationships explicit, the framework serves as a practical mental model to position future projects within the field, supporting clearer comparison across studies and guiding researchers in designing comprehensive systems that seamlessly integrate sensing, modeling, understanding, and intervention to advance routine-aware technologies.

4 Goals of Computing Human Routines in Daily Life

Our review of the literature identified diverse goals for routine computing in daily life, reflecting the different populations, contexts, and scales addressed. Across the studies, we identified four primary goals: supporting aging and disability care, promoting healthy habit formation, enhancing self-reflection through intelligent assistance, and informing community-level understanding through aggregated routines. These goals shaped the behaviors researchers chose to model, the sensors and data modalities emphasized, and the design of supportive technologies. Also, they illustrate how real-world applications extend and operationalize the routine computing dimensions synthesized in Section 3, grounding those abstractions in concrete use contexts. Moreover, we analyzed how the pursuit of each goal positioned routine computing at different levels of impact, ranging from individual well-being to collective social development.

4.1 Supporting Routines in Aging and Disability Care (n=46)

Monitoring human routines in the context of aging and disability care serves two complementary purposes: while disability-focused studies emphasize health-related pattern recognition to support diagnosis and rehabilitation, aging-focused work highlights the tracking of daily routines and risks to promote independence and timely caregiving support.

4.1.1 Monitoring Health-Related Patterns in Disability Contexts (n=11). In disability care, growing research emphasizes the use of routine-oriented sensing to capture daily behaviors as proxies for functional ability, disease trajectories, and rehabilitation progress. A variety of wearable and object-embedded systems have been introduced to track physical activity patterns and movement characteristics, supporting the assessment of motor impairments and neurological conditions [8, 22, 35, 46, 211]. Likewise, ambient and smart home infrastructures integrate multimodal sensing—from cameras and appliance usage to localization technologies—to monitor activities of daily living, detect cognitive decline, and anticipate risks in dementia or physical disability care [6, 30, 70, 98, 125]. Recent studies also indicate that sequence learning and predictive modeling have advantages in recognizing temporal dependencies in patients' daily behaviors, thereby improving the accuracy of activity recognition and the ability to forecast deviations in routine that may signal emerging health concerns [70, 111, 125].

4.1.2 Tracking Daily Routines and Risks in Aging (n=35). A central goal of monitoring routines in aging contexts is to support older adults in maintaining independence while enabling caregivers and family members to understand their daily lives better. Routine-oriented sensing has been used to capture mobility, exercise adherence, and everyday activities, thereby offering valuable insights into functional status, early frailty, and lifestyle changes that may require intervention [5, 51, 55, 76, 86, 89, 118, 141, 178, 198, 203, 207, 223]. Video surveillance approaches allow for non-intrusive monitoring for risk scenarios such as falls or wandering [27, 60, 226]. These insights are particularly valuable in community-dwelling contexts, where sustained monitoring captures gradual behavioral shifts that might otherwise go unnoticed, offering an early window into emerging health concerns [158, 159]. Beyond these settings, routine-oriented systems also support broader caregiving with continuous awareness of older adults' daily well-being and potential care needs [3, 31, 94, 177, 193, 219]. A particularly promising line of work builds on digital twin concepts, integrating comprehensive

data streams to construct dynamic representations of older adults' daily routines, allowing more holistic and proactive monitoring [53, 54, 107, 121].

4.2 Promoting Everyday Health Practices and Specific Habit Patterns (n=55)

Everyday health routines encompass both broad patterns, such as mental well-being and the balance between physical activity and work, as well as more specific lifestyle habits. Together, these routines offer valuable perspectives for understanding and supporting everyday health management.

4.2.1 Indicators of Mental Health States (n=10). Daily routines provide a valuable lens for understanding mental health, as subtle variations in activities, mobility, and physiological signals often reflect underlying emotional states. Studies have shown that wearable and smartphone-based sensing can infer mood, stress, and affective responses from multimodal data such as activity patterns, sleep, diet, and biometrics [26, 102, 183], while longitudinal monitoring further enables the detection of routine changes that may indicate depression or heightened stress levels [108, 126, 137]. Integrating heterogeneous data streams has been found to improve the assessment of mental well-being, offering richer and more reliable indicators than single-source approaches [6, 50, 126, 183, 217]. Also, frameworks have begun linking routine-aware detection to actionable recommendations, highlighting the potential of computing human routines to support proactive strategies for maintaining and enhancing mental health [102, 150].

4.2.2 Equilibrium between Physical Activity and Work (n=31). Balancing physical activity and work-related routines is a key objective in understanding human daily practices. Studies have emphasized the importance of recognizing locomotion, gait, exercise behaviors, and occupational tasks as interconnected elements of health and productivity [14, 16, 68, 106, 133, 167, 169, 221]. In contrast, others highlighted how daily schedules reveal interdependencies between mobility, work patterns, and self-aware activities [37, 73, 179, 206, 216]. In addition, smartphone and multimodal sensing extend this perspective by situating physical activity within broader lifestyle demands, including maintaining exercise plans, balancing work responsibilities with periods of rest, and engaging in leisure or social activities [20, 59, 103, 122, 173, 197]. These insights point to proactive support systems that help individuals maintain an equilibrium between occupational requirements and health-promoting routines, offering opportunities for both performance enhancement and long-term well-being [39, 43, 48, 114, 154, 222].

4.2.3 Fine-Grained Lifestyle Patterns for Health Promotion (n=14). Understanding lifestyle at a fine-grained level is critical for promoting sustainable health behaviors. Studies have demonstrated the value of capturing specific routines such as eating [66, 131, 148], sleep [32, 148, 195], smoking [7], hygiene [142, 151], household tasks [77, 166], and micromotor activities such as handwriting [57] or scratching behaviors [92]. These detailed patterns provide insights into the subtle, often-overlooked aspects of daily life that actually sometimes affect long-term well-being. Complementary approaches utilize smartphones and wearable devices as personal informatics tools to track multidimensional routines and support behavior change through personalized feedback and goal-setting [11, 63]. Together, these works highlight how routine-oriented sensing can reveal the causal links between small-scale habits and broader health outcomes, therefore enabling more targeted interventions and fostering individual awareness and self-regulation [32, 148, 195].

4.3 Designing Adaptive Support through Reflection, Assistance, and Prediction of Human Routines (n=49)

Computing human routines not only enables individuals to reflect on their everyday behaviors but also supports adaptive guidance through context-aware systems. Moreover, it facilitates forward-looking strategies, such as routine prediction and synthetic data generation. These directions illustrate a progression from personal self-awareness toward proactive and assistive technologies.

4.3.1 Fostering Self-Reflection in Living Routines (n=30). Self-reflection in daily life refers to the capacity of individuals to observe and evaluate their own routines, thereby recognizing healthy patterns, imbalances, or disruptions that may affect well-being. In smart home contexts, making routine structures visible, such as meal times, mobility behaviors, electricity usage, or leisure practices, can help individuals better understand their habits and foster healthier lifestyle management [4, 33, 34, 181]. Meanwhile, systems that highlight uniformity or deviations in routines allow users to reflect on adherence and recognize potential risks or lifestyle drifts [138, 156, 174]. Personalized self-reports are triggered based on optimized sampling timeframes to help users reflect on their routines without causing disruption [74, 105]. Beyond domestic settings, workplace-oriented sensing approaches similarly encourage awareness of daily rhythms and vitality [23]. Another strand of research emphasizes interpretability, for example, by representing activities through human-object interactions or semantically structured action sequences, which makes daily behaviors easier to recall, compare, and reason about in ways that support self-awareness [9, 10, 67, 88, 112].

4.3.2 Guiding Human Behaviors with Context-Aware Assistance (n=14). Context-aware assistance aims to guide people in maintaining healthy and productive routines by adapting system responses to ongoing behaviors. Such systems often model the probabilistic sequencing of activities and detect deviations that may indicate risks, allowing timely interventions [62, 210, 214]. Assistance can take the form of alternative recommendations or adaptive feedback loops, where robots, mobile devices, or ambient systems communicate with users and incorporate their input to refine subsequent guidance [95, 135, 136, 210]. More importantly, interventions are not limited to health and safety but also support cultural or religious practices, productivity, and consumption, thereby embedding technology into broader aspects of daily life [52, 123, 163, 201, 212]. Moving beyond passive monitoring to proactive support, these systems exemplify how routine computing can enhance well-being and autonomy through contextually informed assistance [135, 136, 149, 212]. Visually impaired people, for example, benefit from context-aware navigation aids that adapt to their routines and environments, improving mobility and independence [24, 160].

4.3.3 Advancing Routine Prediction and Synthetic Data Generation (n=5). Routine prediction and synthetic data generation have emerged as important research directions in routine modeling. Recent work develops models that capture spatiotemporal patterns of activities, enabling the clustering of ADL sequences and the creation of synthetic variations to test system robustness [65]. Predictive frameworks using motion sensors or IoT devices further support anomaly detection and daily behavior forecasting without relying heavily on manual labeling [18, 130, 139]. Other studies, such as collaborative intent prediction, leverage contextual signals to anticipate real-time user needs, extending routine modeling into digital spheres[185]. Such advances explore how prediction and synthesis can inform human-centered design and research toward adaptive systems.

4.4 From Individual Routines to Population-Informed Insights (n=9)

Population-level analyses of human routines extend the scope of routine computing beyond individual assistance, offering broader perspectives on mobility, social interaction, and collective behavior. By examining aggregated digital traces and behavioral datasets, these studies illuminate how everyday activities scale to inform domains such as public health, education, and smart city management. Large-scale behavioral mining has revealed recurring mobility and lifestyle patterns in urban and community settings, showing how periodic routines can inform city planning and management [39, 47]. Broader datasets, such as video collections and smartphone sensing, capture diverse activities across cultures, allowing more representative benchmarks for routine analysis [128, 224]. Intrinsic connections are explored between individual routines and societal trends, revealing how personal habits aggregate to influence community well-being and public policy [19, 100]. In educational and campus contexts, studies link student routines in learning, consumption, and social life with social influence to predict broader outcomes [46, 83, 126]. At the group level, mobility-based sensing reveals shared routines and collective formations, shedding light on how people coordinate activities, interact socially, and structure daily movements [219].

5 Discussion

Our review in Section 3 synthesizes the sensing patterns, behavioral contexts, cognitive factors, and adaptation mechanisms that characterize routine computing. Section 4 further shows how these systems aim to support health, productivity, and well-being across diverse populations. Building on this foundation, our discussion moves beyond general technical themes to identify explicitly HCI-oriented challenges and opportunities, outlining both the major challenges and the corresponding design directions. In the following discussion, we connect the field's limitations to concrete human-oriented issues—such as interpretability gaps, feedback burden, privacy authorization, and domain-sensitive data practices—and provide design recommendations grounded in user interaction patterns, interface requirements, and human-centered design principles.

5.1 From Activities to Intent: Context and Interpretability

Our findings reveal that routine computing research has made significant progress in detecting low-level events and activities, yet continues to face challenges in bridging the gap from observed behaviors to contextualized interpretations of human intent. Across various sensing modalities, systems often capture *what* action occurred (e.g., walking, cooking, flushing) but fail to determine *why* it occurred or what meaning it holds [22, 67, 141, 167, 181]. This semantic gap leads to interpretability issues: routine deviations can be flagged without clarity on whether they signal benign variation, social influences, or health decline [86, 104, 131, 200]. Similarly, ambiguities arise when identical signals map to multiple possible intentions, such as a person in bed that is asleep, resting, or unwell [67, 80, 119, 141]. Importantly, these challenges extend beyond technical limitations. The lack of contextual grounding prevents systems from supporting users' metacognition, reflection, and self-monitoring of their routines (Section 3.3.5)—processes that are essential for helping people understand not just what they did, but why they behave as they do. Collectively, these limitations constrain the capacity of routine computing systems to move beyond pattern recognition toward meaningful explanations of lifestyle.

To advance toward interpretability, future work should prioritize the integration of richer contextual cues and hierarchical modeling of goals. Technically, combining multi-modal data streams can provide the grounding necessary to differentiate between superficially similar activities [38, 129, 208, 223]. In particular, egocentric sensing [24, 25, 60, 131]-likely to become more common with the adoption of AR glasses-offers a fine-grained first-person context that may help systems disambiguate user intent in ways that stationary or device-centric sensors cannot. Methods that incorporate causal and relational reasoning, such as ontology-based frameworks [67], pattern mining [144], and hierarchical temporal models [75], hold promise for connecting low-level observations with higher-order intentions and values. Moreover, longitudinal analysis of routines over weeks or months could reveal the progression and underlying causes of behavioral changes, rather than isolated deviations [29, 64, 131]. From an interaction perspective, the key is to design transparent and interpretable mechanisms that make these system inferences cognitively accessible to users. Interfaces should reveal how individual actions are combined with contextual cues to infer intentions, allowing users to see and reflect on the causal links between their daily behaviors and system-generated interpretations. For example, if the system flags a drop in morning physical activity, the interface could display contextual information such as shorter-than-usual sleep or adverse weather conditions, helping users understand the inferred behavioral change. This aligns with the goal of context-aware assistance (Section 4.3.2), emphasizing that systems should not only infer intent but also surface explanations in forms that help users understand behavioral goals and potential adjustments, therefore effectively supporting human-centered applications in health, well-being, and daily life.

5.2 Stable yet Adaptive: Balancing Personalization and Generalization in Longitudinal Modeling

Although routine computing systems increasingly capture long-term patterns, they struggle to balance the stability of generalized models with the flexibility required for personalized adaptation. On one hand, models trained on population-level data often fail to account for individual variability in physiology, preferences, and lifestyles [102, 183, 191, 222], limiting their utility in real-world deployment. On the other hand, highly individualized models, such as those based on unique mobility trajectories or home-specific sensor distributions, face challenges in scaling and transferring across users and domains [86, 117, 203, 213, 225]. The dynamic nature of human life further complicates this tension. What constitutes a "normal" routine can shift gradually with age, health changes, or environmental adjustments [67, 70, 86, 175]. Moreover, models are often trained statically on historical data [33, 42, 93], without mechanisms for incremental learning to accommodate evolving habits [28]. Critically, from a user-oriented standpoint, current feedback-driven personalization [14, 48, 61, 193] introduces additional challenges. Users might be required to decide when and how to report routine changes or correct misclassifications, which can impose cognitive load and may interrupt ongoing activities. These issues echo Section 3.3 and Section 3.4, especially the difficulty users face in describing their own routine patterns (Section 3.3.5) and in expressing small deviations in ways that translate into actionable system interventions (Section 3.4.2). Future directions thus lie in developing longitudinal models that are both stable enough to identify enduring regularities and adaptive enough to capture personal evolution, while remaining low-burden and seamlessly integrated into users' everyday lives.

One promising approach is cross-domain fusion, where integrating temporal, spatial, physiological, and social cues can provide resilience to noise while grounding personalization in richer contexts [171, 173, 193, 224]. Transfer learning and domain adaptation methods can also allow models trained on large populations to be efficiently recalibrated for individuals [93, 109, 191, 197]. Meanwhile, interaction design plays a central role in managing feedback-driven personalization [14, 48, 61, 193], which should reduce cognitive burden by offering lightweight feedback channels. Users often struggle to articulate why a routine has changed or what correction is needed, therefore, systems should guide them with example-based prompts, comparative visualizations ("your routine this week vs last week"), and suggestions that can be accepted or rejected with low effort. These mechanisms can transform personalization from a high-effort reporting task into a lightweight negotiation process embedded naturally into everyday interaction.

5.3 Principles into Practice: Design for Privacy and Ethics

Our findings show that privacy and ethical concerns remain a central barrier to the deployment of routine computing systems. Many studies highlight the tension between continuous monitoring for utility and the intrusiveness of the modalities employed, which is further intensified by the fine-grained temporal granularity required for routine computing (Section 3.1). Video and audio-based sensing in homes or workplaces raises concerns about intimate surveillance [91, 165, 173, 183], while wearable and physiological monitoring can expose sensitive biometric and behavioral data when transmitted to external servers or third-party cloud platforms [53, 83, 96]. In addition, when ambient sensors or lifelogging devices are adopted, users often express discomfort with opaque monitoring practices and unclear data handling [33, 57, 74, 95]. Privacy sensitivities are further heightened when sensing is deployed among vulnerable populations in aging and disability care (Section 4.1). These concerns are not only technical but also social. High opt-out rates and refusals to participate in studies manifest how privacy perceptions restrict adoption [74, 101]. Together, these challenges reveal a structural misalignment between the data-intensive requirements of activity recognition and the ethical imperatives of preserving consent and control in everyday environments.

A body of existing research points to multiple strategies for embedding privacy-preserving principles into practice. One approach has been to reduce invasiveness by adopting lens-free and wearable designs that limit the exposure of identifiable data [78, 134, 202]. Other work emphasizes selective or cryptographic data transformations, such as hashed identifiers for location traces [19], anonymization protocols for group mobility [12], or user-controlled masking of sensitive streams [164, 186]. Beyond these system-level safeguards, another key direction is to design interfaces that allow people to understand and control the flow of temporal and contextual data without overwhelming them. For example, progressive disclosure interfaces could show what is being sensed at different levels of granularity. At the same time, preview mechanisms can simulate how routine inferences would change if certain streams were disabled. Such interaction tools allow users to actively shape the privacy boundaries of routine-aware systems rather than passively accept them, thereby enabling the sustainable integration of routine computing into daily life.

5.4 Bridging the Data Chasm: Curation and Enrichment

Despite the abundance of work on routine computing, the field is constrained by a persistent "data chasm." Many studies rely on small and homogeneous datasets that limit ecological validity [52, 108, 129, 172]. Even widely adopted resources such as CASAS [41] provide only partial coverage of real-world variability. Longitudinal data collection remains resource-intensive and difficult to scale [138, 155], while ground-truth labeling is often inconsistent or relies on manual inspection[65, 183, 197]. In-the-wild data introduces further complexity, including noisy, sparse, and unbalanced signals [102, 130, 146], sensor placement inconsistencies [107, 213], and ethical constraints that require data collection from healthy subjects rather than target groups [22, 46]. These constraints not only hinder generalization across devices and environments [39, 93], but also obscure the detection of rare or anomalous events that are often artificially induced rather than naturally observed [112, 210]. Therefore, these challenges underscore the gap between experimental feasibility and the demands of robust and scalable routine computing, particularly in supporting goals such as translating individual routines into population-informed insights (Section 4.4).

In response, recent work points toward a layered set of strategies for bridging this chasm. One orientation is toward multi-dataset training and cross-source integration, which can mitigate modality-specific biases and support broader generalization[19, 93]. While real-world data remains scarce, synthetic dataset generation and co-occurrence analysis are utilized. Innovations in labeling protocols, such as standardized schemes[5] or sophisticated self-reporting applications[197], suggest pathways to improve consistency in ground truth acquisition. Apart from algorithmic advances[102, 130], future work should also emphasize human-group-sensitive curation principles, such as collecting data from clinically or socially relevant populations [22, 46]. There is also an opportunity for HCI to support more sustainable practices by designing tools that help participants contribute longitudinal data with less burden. Examples include self-report aids, micro-interaction logging tools, or interfaces that summarize routine changes and allow users to validate or annotate them retroactively. Moreover, egocentric life-logging systems can be conceptualized to gain higher-quality personal routine data. Such platforms can be designed by fusing various sensing modalities with privacy-aware mechanisms to protect sensitive information and ensure transparency. All these directions suggest a shift from isolated datasets and ad hoc evaluation toward an integrated and enriched resource ecosystem. Addressing this data chasm is not merely a technical prerequisite but a foundational step for enabling routine computing to move from controlled prototypes to sustainable, real-world systems that genuinely serve human users.

6 Limitations

A limitation of this review is that our database scope was restricted to ACM and IEEE venues, which historically anchor routine computing through HCI, ubiquitous computing, sensing, and AI research. While preliminary scoping of Springer, ScienceDirect, and NIH yielded primarily medical or clinically oriented monitoring studies with limited computational routine modeling relevance, future work may broaden coverage across these sources to capture cross-disciplinary perspectives and further strengthen comprehensiveness.

7 Conclusion

This review synthesized 203 studies to chart the evolving landscape of routine computing for daily life. We identified four key dimensions that characterize how routines are sensed and understood—temporal granularity, interactional contexts, cognitive-psychological factors, and the dynamics of variability and disruption. We further organized the goals of routine computing into domains spanning support for aging and disability care, the promotion of health practices and habits, adaptive assistance through reflection and prediction, and the extension from individual routines to population-level insights. Informed by these results, our discussion is structured around two overarching themes: persistent challenges and future prospects, including enhancing interpretability of routine models, balancing personalization with generalizability in longitudinal settings, embedding privacy and ethical principles into system design, and advancing data curation and enrichment practices. Together, these directions outline a research agenda that positions routine computing as a critical frontier for human-centered HCI innovation in everyday life.

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References

[1] Gregory D. Abowd, Anind K. Dey, Peter J. Brown, Nigel Davies, Mark Smith, and Pete Steggles. 1999. Towards a Better Understanding of Context and Context-Awareness. In *Proceedings of the 1st International Symposium on Handheld and Ubiquitous Computing* (Karlsruhe, Germany) *(HUC '99)*. Springer-Verlag, Berlin, Heidelberg, 304–307.

[2] Fateme Akbari and Kamran Sartipi. 2024. A Model for Detecting Abnormality in Activities of Daily Living Sequences Using Inverse Reinforcement Learning. In *Proceedings of the 39th ACM/SIGAPP Symposium on Applied Computing*. 1031–1033.

[3] José Alcalá, Oliver Parson, and Alex Rogers. 2015. Detecting anomalies in activities of daily living of elderly residents via energy disaggregation and cox processes. In *Proceedings of the 2nd ACM international conference on embedded systems for energy-efficient built environments*. 225–234.

[4] H. Karamath Ali and D. I. George Amalarethinam. 2014. Activity Recognition with Fuzzy Finite Automata. In *2014 World Congress on Computing and Communication Technologies*. 222–227. doi:10.1109/WCCCT.2014.34

[5] Sayeda Shamma Alia and Paula Lago. 2024. Daily Routine Recognition from Longitudinal, Real-Life Wearable Sensor Data for the Elderly. In *2024 International Conference on Activity and Behavior Computing (ABC)*. 1–9. doi:10.1109/ABC61795.2024.10651644

[6] Oliver Amft and Clemens Lombriser. 2011. Modelling of distributed activity recognition in the home environment. In *2011 Annual International Conference of the IEEE Engineering in Medicine and Biology Society*. 1781–1784. doi:10.1109/IEMBS.2011.6090508

[7] Edwin Valarezo Añazco, Patricio Rivera Lopez, Sangmin Lee, Kyungmin Byun, and Tae-Seong Kim. 2018. Smoking activity recognition using a single wrist IMU and deep learning light. In *Proceedings of the 2nd international conference on digital signal processing*. 48–51.

[8] Riku Arakawa, Karan Ahuja, Kristie Mak, Gwendolyn Thompson, Sam Shaaban, Oliver Lindhiem, and Mayank Goel. 2023. Lemurdx: using unconstrained passive sensing for an objective measurement of hyperactivity in children with no parent input. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 2 (2023), 1–23.

[9] Yutaka Arakawa, Keiichi Yasumoto, Krita Pattamasiriwat, and Teruhiro Mizumoto. 2017. Improving recognition accuracy for activities of daily living by adding time and area related features. In *2017 Tenth International Conference on Mobile Computing and Ubiquitous Network (ICMU)*. 1–6. doi:10.23919/ICMU.2017.8330104

[10] Joan Aranda and Manuel Vinagre. 2016. Anticipating human activities from object interaction cues. In *2016 25th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)*. IEEE, 58–63.

[11] Hania Aslam, Hamid Mukhtar, Farhana Seemi, and Djamel Belaïd. 2016. Harnessing Smartphones as a Personal Informatics Tool towards Self-Awareness and Behavior Improvement. In *2016 IEEE 14th Intl Conf on Dependable, Autonomic and Secure Computing, 14th Intl Conf on Pervasive Intelligence and Computing, 2nd Intl Conf on Big Data Intelligence and Computing and Cyber Science and Technology Congress(DASC/PiCom/DataCom/CyberSciTech)*. 467–474. doi:10.1109/DASC-PICom-DataCom-CyberSciTec.2016.92

[12] Akanksha Atrey, Camellia Zakaria, Rajesh Balan, and Prashant Shenoy. 2024. W4-Groups: Modeling the Who, What, When and Where of Group Behavior via Mobility Sensing. *Proceedings of the ACM on Human-Computer Interaction* 8, CSCW1 (2024), 1–29.

[13] Muhammad Awais, Sabato Mellone, and Lorenzo Chiari. 2015. Physical activity classification meets daily life: Review on existing methodologies and open challenges. In *2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. 5050–5053. doi:10.1109/EMBC.2015.7319526

[14] Sumair Aziz, Muhammad Umar Khan, Ahmad Zahoor, and Syed Zohaib Hassan Naqvi. 2020. Intelligent System for Human Context Recognition. In *2020*

International Conference on Computing and Information Technology (ICCIT-1441). 1–5. doi:10.1109/ICCIT-144147971.2020.9213805

[15] Tunç Aşuroğlu. 2022. Complex Human Activity Recognition Using a Local Weighted Approach. *IEEE Access* 10 (2022), 101207–101219. doi:10.1109/ACCESS.2022.3208686

[16] Sheikh Badar ud din Tahir, Ahmad Jalal, and Kibum Kim. 2021. Daily life Log Recognition based on Automatic Features for Health care Physical Exercise via IMU Sensors. In *2021 International Bhurban Conference on Applied Sciences and Technologies (IBCAST)*. 494–499. doi:10.1109/IBCAST51254.2021.9393204

[17] Nikola Banovic, Tofi Buzali, Fanny Chevalier, Jennifer Mankoff, and Anind K Dey. 2016. Modeling and understanding human routine behavior. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. 248–260.

[18] Nikola Banovic, Anqi Wang, Yanfeng Jin, Christie Chang, Julian Ramos, Anind Dey, and Jennifer Mankoff. 2017. Leveraging human routine models to detect and generate human behaviors. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 6683–6694.

[19] Xuan Bao, Neil Zhenqiang Gong, Bing Hu, Yilin Shen, and Hongxia Jin. 2014. Connect the dots by understanding user status and transitions. In *Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing: Adjunct Publication*. 361–366.

[20] Erin K Barrett, Cameron M Fard, Hannah N Katinas, Charles V Moens, Lauren E Perry, Blake E Ruddy, Shalin D Shah, Ian S Tucker, Tucker J Wilson, Mark Rucker, et al. 2020. Mobile sensing: Leveraging machine learning for efficient human behavior modeling. In *2020 Systems and Information Engineering Design Symposium (SIEDS)*. IEEE, 1–7.

[21] Mouazma Batool and Ahmad Jalal. 2023. Telemonitoring of Daily Activities Based on Multi-Sensors Data Fusion. In *2023 20th International Bhurban Conference on Applied Sciences and Technology (IBCAST)*. 184–189. doi:10.1109/IBCAST59916.2023.10712809

[22] Maxence Bobin, Hamdi Amroun, Mehdi Boukalle, Margarita Anastassova, and Mehdi Ammi. 2018. Smart Cup to Monitor Stroke Patients Activities During Everyday Life. In *2018 IEEE International Conference on Internet of Things (iThings) and IEEE Green Computing and Communications (GreenCom) and IEEE Cyber, Physical and Social Computing (CPSCom) and IEEE Smart Data (SmartData)*. 189–195. doi:10.1109/Cybermatics_2018.2018.00062

[23] Hans Brombacher, Steven Houben, and Steven Vos. 2022. Sensorbadge: an exploratory study of an ego-centric wearable sensor system for healthy office environments. In *Proceedings of the 2022 ACM Designing Interactive Systems Conference*. 1863–1877.

[24] Andreas Bulling, Christian Weichel, and Hans Gellersen. 2013. EyeContext: recognition of high-level contextual cues from human visual behaviour. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Paris, France) *(CHI '13)*. Association for Computing Machinery, New York, NY, USA, 305–308. doi:10.1145/2470654.2470697

[25] Jeffrey Byrne, Greg Castañón, Zhongheng Li, and Gil Ettinger. 2023. Fine-grained Activities of People Worldwide. In *2023 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. 3307–3318. doi:10.1109/WACV56688.2023.00332

[26] Yekta Said Can and Elisabeth André. 2023. Performance exploration of rnn variants for recognizing daily life stress levels by using multimodal physiological signals. In *Proceedings of the 25th international conference on multimodal interaction*. 481–487.

[27] Yuanyuan Cao, Linmi Tao, and Guangyou Xu. 2009. An Event-driven Context Model in Elderly Health Monitoring. In *2009 Symposia and Workshops on Ubiquitous, Autonomic and Trusted Computing*. 120–124. doi:10.1109/UIC-ATC.2009.47

[28] Berardina De Carolis, Stefano Ferilli, and Domenico Redavid. 2015. Incremental learning of daily routines as workflows in a smart home environment. *ACM Transactions on Interactive Intelligent Systems (TiiS)* 4, 4 (2015), 1–23.

[29] Loïc Caroux, Charles Consel, Lucile Dupuy, and Hélène Sauzéon. 2014. Verification of daily activities of older adults: a simple, non-intrusive, low-cost approach. In *Proceedings of the 16th international ACM SIGACCESS conference on Computers & accessibility*. 43–50.

[30] C. Chalmers, P. Fergus, C. Aday Curbelo Montanez, S. Sikdar, F. Ball, and B. Kendall. 2022. Detecting Activities of Daily Living and Routine Behaviours in Dementia Patients Living Alone Using Smart Meter Load Disaggregation. *IEEE Transactions on Emerging Topics in Computing* 10, 1 (2022), 157–169. doi:10.1109/TETC.2020.2993177

[31] Rhian Chambers and Muhammad Fahim. 2022. Healthy Aging: A Proactive Model to Prevent Self-neglecting Behavior in Smart Homes. In *2022 IEEE International Conference on E-health Networking, Application & Services (HealthCom)*. 173–178. doi:10.1109/HealthCom54947.2022.9982760

[32] Liqiong Chang, Jiaqi Lu, Ju Wang, Xiaojiang Chen, Dingyi Fang, Zhanyong Tang, Petteri Nurmi, and Zheng Wang. 2018. SleepGuard: Capturing rich sleep information using smartwatch sensing data. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 2, 3 (2018), 1–34.

[33] Sudarshan S. Chawathe. 2018. Recognizing Activities of Daily Living Using Binary Sensors. In *2018 4th International Conference on Universal Village (UV)*. 1–6. doi:10.1109/UV.2018.8642134

[34] Li Chen and William K. Cheung. 2014. Recovering Human Mobility Flow Models and Daily Routine Patterns in a Smart Environment. In *2014 IEEE International Conference on Data Mining Workshop*. 541–548. doi:10.1109/ICDMW.2014.155

[35] Yineng Chen, Feng Tian, Xinda Zeng, Yue He, Chaoci Jiang, Xiaolong" Luke" Zhang, Yicheng Zhu, and Hongan Wang. 2014. UbiSpoon: pervasive monitoring of nervous system diseases through daily life. In *CHI'14 Extended Abstracts on Human Factors in Computing Systems*. 2395–2400.

[36] Zhuolong Chen, Yubin Zhao, and Cheng-Zhong Xu. 2025. Lightweight HAR Scheme for Rapid Environment Adaption Based on AIoT WiFi Sensing Chips. *IEEE Internet of Things Journal* 12, 13 (2025), 25908–25921. doi:10.1109/JIOT.2025.3561040

[37] Heng-Tze Cheng, Feng-Tso Sun, Martin Griss, Paul Davis, Jianguo Li, and Di You. 2013. NuActiv: recognizing unseen new activities using semantic attribute-based learning. In *Proceeding of the 11th Annual International Conference on Mobile Systems, Applications, and Services* (Taipei, Taiwan) *(MobiSys '13)*. Association for Computing Machinery, New York, NY, USA, 361–374. doi:10.1145/2462456.2464438

[38] Emil Stefan Chifu, Viorica Rozina Chifu, Cristina Bianca Pop, Alin Vlad, and Ioan Salomie. 2018. Machine Learning Based Technique for Detecting Daily Routine and Deviations. In *2018 IEEE 14th International Conference on Intelligent Computer Communication and Processing (ICCP)*. 183–189. doi:10.1109/ICCP.2018.8516598

[39] Gabriela Ciortuz, Marcin Grzegorzek, and Sebastian Fudickar. 2023. Effects of time-series data pre-processing on the transformer-based classification of activities from smart glasses. In *Proceedings of the 8th international Workshop on Sensor-Based Activity Recognition and Artificial Intelligence*. 1–4.

[40] Andy Clark. 1981. *Being There: Putting Brain, Body, and World Together Again*. MIT Press.

[41] Diane J Cook, Aaron S Crandall, Brian L Thomas, and Narayanan C Krishnan. 2012. CASAS: A smart home in a box. *Computer* 46, 7 (2012), 62–69.

[42] Nicholas J. Cooney, Karna Prasanna Joshi, and Atul S. Minhas. 2018. A Wearable Internet of Things Based System with Edge Computing for Real-Time Human Activity Tracking. In *2018 5th Asia-Pacific World Congress on Computer Science and Engineering (APWC on CSE)*. 26–31. doi:10.1109/APWConCSE.2018.00013

[43] Pablo Fernandez de Dios, Paul Wai Hing Chung, and Qinggang Meng. 2014. Landmark-Based Methods for Temporal Alignment of Human Motions. *IEEE Computational Intelligence Magazine* 9, 2 (2014), 29–37. doi:10.1109/MCI.2014.2307223

[44] Edward L. Deci and Richard M. Ryan. 1985. *Intrinsic Motivation and Self-Determination in Human Behavior*. doi:10.1007/978-1-4899-2271-7

[45] Milan Deumer, Moid Sandhu, Sara Khalifa, Brano Kusy, Kai Geissdoerfer, Marco Zimmerling, and Raja Jurdak. 2023. A battery-free wearable system for on-device human activity recognition using kinetic energy harvesting. In *EWSN'23: Proceedings of the 2023 International Conference on embedded Wireless Systems and Networks*. Association for Computing Machinery (ACM), 163–169.

[46] U Sobiha Devi, Madhavan Bharanidivya, and Samiappan Dhanalakshmi. 2024. Wearable Device for Proactive Diagnosis and Monitoring Daily Activities in Parkinson's Disease Patients. In *2024 IEEE 21st India Council International Conference (INDICON)*. IEEE, 1–5.

[47] Yong Ding, Julio Borges, Martin A Neumann, and Michael Beigl. 2015. Sequential pattern mining—A study to understand daily activity patterns for load forecasting enhancement. In *2015 IEEE First International Smart Cities Conference (ISC2)*. IEEE, 1–6.

[48] Aysegül Dogangün, Michael Schwarz, Katharina Kloppenborg, and Robert Le. 2017. An approach to improve physical activity by generating individual implementation intentions. In *Adjunct Publication of the 25th Conference on User Modeling, Adaptation and Personalization*. 370–375.

[49] Paul Dourish. 2001. *Where the Action Is: The Foundations of Embodied Interaction*. The MIT Press. doi:10.7551/mitpress/7221.001.0001

[50] Zhanwei Du, Yongjian Yang, Chuang Ma, and Bo Yang. 2014. Negative energy detector using cellphone bluetooth and contact list. In *Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing: Adjunct Publication*. 43–46.

[51] Andreas Ejupi, Matthew Brodie, Stephen R Lord, Janneke Annegarn, Stephen J Redmond, and Kim Delbaere. 2016. Wavelet-based sit-to-stand detection and assessment of fall risk in older people using a wearable pendant device. *IEEE Transactions on Biomedical Engineering* 64, 7 (2016), 1602–1607.

[52] Khaled Eskaf, Walid Mohamed Aly, and Alyaa Aly. 2016. Aggregated Activity Recognition Using Smart Devices. In *2016 3rd International Conference on Soft Computing & Machine Intelligence (ISCMI)*. 214–218. doi:10.1109/ISCMI.2016.52

[53] Muhammad Fahim, Vishal Sharma, Ruth Hunter, and Trung Q. Duong. 2023. Healthy Aging: A Deep Meta-Class Sequence Model to Integrate Intelligence in Digital Twin. *IEEE Journal of Translational Engineering in Health and Medicine* 11 (2023), 330–340. doi:10.1109/JTEHM.2023.3274357

[54] Lin Fan, Zhongmin Wang, and Hai Wang. 2013. Human Activity Recognition Model Based on Decision Tree. In *2013 International Conference on Advanced Cloud and Big Data*. 64–68. doi:10.1109/CBD.2013.19

[55] Umer Fareed. 2015. Smartphone sensor fusion based activity recognition system for elderly healthcare. In *Proceedings of the 2015 workshop on pervasive wireless healthcare*. 29–34.

[56] Katayoun Farrahi and Daniel Gatica-Perez. 2008. What did you do today? discovering daily routines from large-scale mobile data. In *Proceedings of the 16th ACM International Conference on Multimedia* (Vancouver, British Columbia, Canada) *(MM '08)*. Association for Computing Machinery, New York, NY, USA, 849–852. doi:10.1145/1459359.1459503

[57] Davide Di Febbo, Francesca Lunardini, Milad Malavolti, Alessandra Pedrocchi, N. Alberto Borghese, and Simona Ferrante. 2020. IoT ink pen for ecological monitoring of daily life handwriting. In *2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*. 5749–5752. doi:10.1109/EMBC44109.2020.9175999

[58] John Flavell. 1979. Metacognition and Cognitive Monitoring: A New Area of Cognitive-Developmental Inquiry. *American Psychologist* 34 (10 1979), 906–911. doi:10.1037/0003-066X.34.10.906

[59] Anthony Fleury, Norbert Noury, and Michel Vacher. 2010. Introducing knowledge in the process of supervised classification of activities of Daily Living in Health Smart Homes. In *The 12th IEEE International Conference on e-Health Networking, Applications and Services*. 322–329. doi:10.1109/HEALTH.2010.5556549

[60] Homa Foroughi, Baharak Shakeri Aski, and Hamidreza Pourreza. 2008. Intelligent video surveillance for monitoring fall detection of elderly in home environments. In *2008 11th International Conference on Computer and Information Technology*. 219–224. doi:10.1109/ICCITECHN.2008.4803020

[61] Dany Fortin-Simard, Sebastien Gaboury, Bruno Bouchard, and Abdenour Bouzouane. 2015. Frequent pattern clustering for ADLs recognition in smart environments. In *Proceedings of the 8th ACM International Conference on PErvasive Technologies Related to Assistive Environments*. 1–4.

[62] Kevin Fouquet, Gregory Faraut, and Jean-Jacques Lesage. 2020. Life Habits Modeling with Stochastic Timed Automata in Ambient Assisted Living. In *2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. 2740–2745. doi:10.1109/SMC42975.2020.9282900

[63] Saixiong Gan. 2024. Integrating Digital Media and Computer Technology into Designing Products for Children and Older Adults. In *Proceedings of the 2024 International Conference on Big Data Mining and Information Processing*. 326–331.

[64] Shan Gao and Ah-Hwee Tan. 2014. User daily activity pattern learning: A multi-memory modeling approach. In *2014 International Joint Conference on Neural Networks (IJCNN)*. 1542–1548. doi:10.1109/IJCNN.2014.6889908

[65] Shan Gao, Ah-Hwee Tan, and Rossi Setchi. 2021. Learning ADL Daily Routines with Spatiotemporal Neural Networks. *IEEE Transactions on Knowledge and Data Engineering* 33, 1 (2021), 143–153. doi:10.1109/TKDE.2019.2924623

[66] Daniel Gatica-Perez, Joan-Isaac Biel, David Labbe, and Nathalie Martin. 2019. Discovering eating routines in context with a smartphone app. In *Adjunct Proceedings of the 2019 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2019 ACM International Symposium on Wearable Computers*. 422–429.

[67] Geethalekshmy, V S Abishek, and Aditya Nair. 2023. Human Activity Recognition using Ontology. In *2023 4th IEEE Global Conference for Advancement in Technology (GCAT)*. 1–7. doi:10.1109/GCAT59970.2023.10353416

[68] Yazeed Yasin Ghadi, Madiha Javeed, Mohammed Alarfaj, Tamara Al Shloul, Suliman A. Alsuhibany, Ahmad Jalal, Shaharyar Kamal, and Dong-Seong Kim. 2022. MS-DLD: Multi-Sensors Based Daily Locomotion Detection via Kinematic-Static Energy and Body-Specific HMMs. *IEEE Access* 10 (2022), 23964–23979. doi:10.1109/ACCESS.2022.3154775

[69] James J. Gibson. 1979. *The Ecological Approach to Visual Perception*. Houghton Mifflin, Boston, MA. Reprinted by Psychology Press, 2014.

[70] Ismael Miranda Gordo, Ana Jiménez Martín, David Gualda Gómez, Juan Jesús García Domínguez, and Sara García de Villa. 2020. Symbolic localization of institutionalized patients for detection of daily living activities. In *2020 IEEE International Symposium on Medical Measurements and Applications (MeMeA)*. IEEE, 1–5.

[71] Rúben Gouveia, Sergio Barros, and Evangelos Karapanos. 2014. Understanding users' disengagement with wearable activity trackers. In *Proceedings of the 2014 workshops on advances in computer entertainment conference*. 1–3.

[72] Yulong Gu, Mengjia Feng, Yuan Yao, Weidong Liu, and Jiaxing Song. 2016. We Know What You Are Doing or Going to Do: Towards Accurate Human Activities Sensing. In *2016 25th International Conference on Computer Communication and Networks (ICCCN)*. 1–9. doi:10.1109/ICCCN.2016.7568597

[73] Yulong Gu, Weidong Liu, and Jiaxing Song. 2015. Can Activities of Human Daily Life be Recognized and Predicted?. In *2015 IEEE/WIC/ACM International Conference on Web Intelligence and Intelligent Agent Technology (WI-IAT)*, Vol. 1. 127–132. doi:10.1109/WI-IAT.2015.61

[74] Alejandro Sanchez Guinea, Andrey Boytsov, Ludovic Mouline, and Yves Le Traon. 2019. Smart discovery of periodic-frequent human routines for home automation. In *Proceedings of the 16th EAI International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services*. 268–277.

[75] Vinayak Gupta and Srikanta Bedathur. 2024. Tapestry of Time and Actions: Modeling Human Activity Sequences Using Temporal Point Process Flows. *ACM Transactions on Intelligent Systems and Technology* 15, 3 (2024), 1–27.

[76] Zahra Hajihashemi and Mihail Popescu. 2014. A new illness recognition framework using frequent temporal pattern mining. In *Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing: Adjunct Publication*. 1241–1247.

[77] Derek Hao Hu, Sinno Jialin Pan, Vincent Wenchen Zheng, Nathan Nan Liu, and Qiang Yang. 2008. Real world activity recognition with multiple goals. In *Proceedings of the 10th International Conference on Ubiquitous Computing* (Seoul, Korea) *(UbiComp '08)*. Association for Computing Machinery, New York, NY, USA, 30–39. doi:10.1145/1409635.1409640

[78] Michael Hardegger, Daniel Roggen, Alberto Calatroni, and Gerhard Tröster. 2016. S-SMART: A unified bayesian framework for simultaneous semantic mapping, activity recognition, and tracking. *ACM Transactions on Intelligent Systems and Technology (TIST)* 7, 3 (2016), 1–28.

[79] Netzahualcoyotl Hernandez, Muhammad Asif Razzaq, Chris Nugent, Ian McChesney, and Shuai Zhang. 2018. Transfer learning and data fusion approach to recognize activities of daily life. In *Proceedings of the 12th EAI International Conference on Pervasive Computing Technologies for Healthcare*. 227–231.

[80] Yoshiki Honda, Hirozumi Yamaguchi, and Teruo Higashino. 2019. Daily Activity Recognition based on Markov Logic Network for Elderly Monitoring. In *2019 16th IEEE Annual Consumer Communications & Networking Conference (CCNC)*. 1–6. doi:10.1109/CCNC.2019.8651817

[81] Aadel Howedi, Ahmad Lotfi, and Amir Pourabdollah. 2020. A multi-scale fuzzy entropy measure for anomaly detection in activities of daily living. In *Proceedings of the 13th ACM international conference on PErvasive technologies related to assistive environments*. 1–8.

[82] Peng-Ju Hsieh, Yen-Liang Lin, Yu-Hsiu Chen, and Winston Hsu. 2016. Egocentric activity recognition by leveraging multiple mid-level representations. In *2016 IEEE International Conference on Multimedia and Expo (ICME)*. 1–6. doi:10.1109/ICME.2016.7552937

[83] Yifei Huang, Guo Chen, Jilan Xu, Mingfang Zhang, Lijin Yang, Baoqi Pei, Hongjie Zhang, Lu Dong, Yali Wang, Limin Wang, and Yu Qiao. 2024. EgoExoLearn: A Dataset for Bridging Asynchronous Ego- and Exo-centric View of Procedural Activities in Real World. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 22072–22086. doi:10.1109/CVPR52733.2024.02084

[84] Dina Hussein and Ganapati Bhat. 2024. SensorGAN: A novel data recovery approach for wearable human activity recognition. *ACM Transactions on Embedded Computing Systems* 23, 3 (2024), 1–28.

[85] Edwin Hutchins. 1995. *Cognition in the Wild*. MIT Press.

[86] Omar A Ibrahim, James Keller, and Mihail Popescu. 2017. Context preserving representation of daily activities in elder care. In *2017 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*. 547–551. doi:10.1109/BIBM.2017.8217707

[87] Jose Antonio Iglesias, Plamen Angelov, Agapito Ledezma, and Araceli Sanchis. 2011. Evolving human activity classifier from sensor streams. In *2011 IEEE Workshop on Evolving and Adaptive Intelligent Systems (EAIS)*. 139–146. doi:10.1109/EAIS.2011.5945921

[88] Haruya Ishikawa, Yuchi Ishikawa, Shuichi Akizuki, and Yoshimitsu Aoki. 2019. Human-Object Maps for Daily Activity Recognition. In *2019 16th International Conference on Machine Vision Applications (MVA)*. 1–6. doi:10.23919/MVA.2019.8757896

[89] Walaa N. Ismail, Mohammad Mehedi Hassan, and Hessah A. Alsalamah. 2019. Context-Enriched Regular Human Behavioral Pattern Detection From Body Sensors Data. *IEEE Access* 7 (2019), 33834–33850. doi:10.1109/ACCESS.2019.2904122

[90] Ankita Jain and Vivek Kanhangad. 2018. Human Activity Classification in Smartphones Using Accelerometer and Gyroscope Sensors. *IEEE Sensors Journal* 18, 3 (2018), 1169–1177. doi:10.1109/JSEN.2017.2782492

[91] Ahmad Jalal and S. Kamal. 2014. Real-time life logging via a depth silhouette-based human activity recognition system for smart home services. In *2014 11th IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)*. 74–80. doi:10.1109/AVSS.2014.6918647

[92] Zygimantas Jocys, Arash Pouryazdan, and Daniel Roggen. 2020. Multimodal fusion of IMUs and EPS body-worn sensors for scratch recognition. In *Proceedings of the 14th EAI International Conference on Pervasive Computing Technologies for Healthcare*. 325–335.

[93] Shreehar Joshi and Eman Abdelfattah. 2021. Deep Neural Networks for Time Series Classification in Human Activity Recognition. In *2021 IEEE 12th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON)*. 0559–0566. doi:10.1109/IEMCON53756.2021.9623228

[94] Ossi Kaltiokallio, Maurizio Bocca, and Neal Patwari. 2012. Follow @grandma: Long-term device-free localization for residential monitoring. In *37th Annual IEEE Conference on Local Computer Networks - Workshops*. 991–998. doi:10.1109/LCNW.2012.6424092

[95] Tetsuya Kaneko, Reona Gomi, Eri Sato-Shimokawara, and Toru Yamaguchi. 2014. Acquisition of the human characteristics through dialogue with a robot at

home. In *2014 International Symposium on Micro-NanoMechatronics and Human Science (MHS)*. IEEE, 1–4.

[96] E. Kantoch, P. Augustyniak, M. Markiewicz, and D. Prusak. 2014. Monitoring activities of daily living based on wearable wireless body sensor network. In *2014 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*. 586–589. doi:10.1109/EMBC.2014.6943659

[97] Peng Yua Kao, Yan-Jing Lei, Chia-Hao Chang, Chu-Song Chen, Ming-Sui Lee, and Yi-Ping Hung. 2021. Activity Recognition Using First-Person-View Cameras Based on Sparse Optical Flows. In *2020 25th International Conference on Pattern Recognition (ICPR)*. 81–86. doi:10.1109/ICPR48806.2021.9412330

[98] Anastasios Karakostas, Ioulietta Lazarou, Georgios Meditskos, Thanos G. Stavropoulos, Ioannis Kompatsiaris, and Magda Tsolaki. 2015. Sensor-based in-home monitoring of people with dementia using remote web technologies. In *2015 International Conference on Interactive Mobile Communication Technologies and Learning (IMCL)*. 353–357. doi:10.1109/IMCTL.2015.7359618

[99] Sidney Katz, Amasa B. Ford, Roland W. Moskowitz, Beverly A. Jackson, and Marjorie W. Jaffe. 1963. Studies of Illness in the Aged: The Index of ADL: A Standardized Measure of Biological and Psychosocial Function. *JAMA* 185, 12 (09 1963), 914–919. doi:10.1001/jama.1963.03060120024016

[100] Daniel Kelly, Barry Smyth, and Brian Caulfield. 2013. Uncovering Measurements of Social and Demographic Behavior From Smartphone Location Data. *IEEE Transactions on Human-Machine Systems* 43, 2 (2013), 188–198. doi:10.1109/TSMC.2013.2238926

[101] Mohammed Khwaja, Sumer S Vaid, Sara Zannone, Gabriella M Harari, A Aldo Faisal, and Aleksandar Matic. 2019. Modeling personality vs. modeling personalidad: In-the-wild mobile data analysis in five countries suggests cultural impact on personality models. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 3, 3 (2019), 1–24.

[102] Byung Hyung Kim, Sungho Jo, and Sunghee Choi. 2022. ALIS: Learning Affective Causality Behind Daily Activities From a Wearable Life-Log System. *IEEE Transactions on Cybernetics* 52, 12 (2022), 13212–13224. doi:10.1109/TCYB.2021.3106638

[103] Hyoungnyoun Kim, Ig-Jae Kim, Hyoung-gon Kim, and Ji-Hyung Park. 2008. Adaptive Modeling of a User's Daily Life with a Wearable Sensor Network. In *2008 Tenth IEEE International Symposium on Multimedia*. 527–532. doi:10.1109/ISM.2008.56

[104] Kijun Kim, Noboru Koshizuka, and Shota Narisawa. 2023. Application of Smart Meters for Detecting Human Activity: Concept and Case Study. In *2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. 5188–5193. doi:10.1109/SMC53992.2023.10393963

[105] Predrag Klasnja, Beverly L. Harrison, Louis LeGrand, Anthony LaMarca, Jon Froehlich, and Scott E. Hudson. 2008. Using wearable sensors and real time inference to understand human recall of routine activities. In *Proceedings of the 10th International Conference on Ubiquitous Computing* (Seoul, Korea) *(UbiComp '08)*. Association for Computing Machinery, New York, NY, USA, 154–163. doi:10.1145/1409635.1409656

[106] Anna S. Klingenberg, Valentin Purrucker, Willi Schüler, Nagarajan Ganapathy, Nicolai Spicher, and Thomas M. Deserno. 2022. Human activity recognition from textile electrocardiograms. In *2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*. 3434–3437. doi:10.1109/EMBC48229.2022.9871210

[107] Toru Kobayashi, Kazuki Fukae, Tetsuo Imai, and Kenichi Arai. 2022. Digital twin agent for super-aged society. In *2022 IEEE International Conference on Consumer Electronics (ICCE)*. IEEE, 1–6.

[108] Anita de Mello Koch, Nicholas Kastanos, and Vered Aharonson. 2020. Monitoring Routine Changes From Daily Smartphone Usage. In *2020 IEEE International Conference on Human-Machine Systems (ICHMS)*. 1–6. doi:10.1109/ICHMS49158.2020.9209417

[109] Quan Kong, Ziming Wu, Ziwei Deng, Martin Klinkigt, Bin Tong, and Tomokazu Murakami. 2019. MMAct: A Large-Scale Dataset for Cross Modal Human Action Understanding. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*. 8657–8666. doi:10.1109/ICCV.2019.00875

[110] Kundan Krishna, Deepali Jain, Sanket V Mehta, and Sunav Choudhary. 2018. An lstm based system for prediction of human activities with durations. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 1, 4 (2018), 1–31.

[111] Thomas Kuebert, Henning Puder, and Heinz Koeppl. 2021. Improving Daily Routine Recognition in Hearing Aids Using Sequence Learning. *IEEE Access* 9 (2021), 93237–93247. doi:10.1109/ACCESS.2021.3092763

[112] Hilde Kuehne, Ali Arslan, and Thomas Serre. 2014. The Language of Actions: Recovering the Syntax and Semantics of Goal-Directed Human Activities. In *2014 IEEE Conference on Computer Vision and Pattern Recognition*. 780–787. doi:10.1109/CVPR.2014.105

[113] Jeremy Kulchyk and Ali Etemad. 2019. Activity Recognition with Wearable Accelerometers using Deep Convolutional Neural Network and the Effect of Sensor Placement. In *2019 IEEE SENSORS*. 1–4. doi:10.1109/SENSORS43011.2019.8956668

[114] Martin Kusserow, Oliver Amft, and Gerhard Tröster. 2013. Modeling arousal phases in daily living using wearable sensors. *IEEE Transactions on Affective Computing* 4, 1 (2013), 93–105. doi:10.1109/T-AFFC.2012.37

[115] M Powell Lawton and Elaine M Brody. 1969. Assessment of older people: self-maintaining and instrumental activities of daily living. *The gerontologist* (1969).

[116] Victor R Lee, Joel R Drake, Ryan Cain, and Jeffrey Thayne. 2015. Opportunistic uses of the traditional school day through student examination of Fitbit activity tracker data. In *Proceedings of the 14th International conference on interaction design and children*. 209–218.

[117] Airui Li, Qiujian Lv, Yuanyuan Qiao, and Jie Yang. 2016. Improving mobility prediction performance with state based prediction method when the user departs from routine. In *2016 IEEE International Conference on Big Data Analysis (ICBDA)*. 1–7. doi:10.1109/ICBDA.2016.7509835

[118] Chen Li, William K. Cheung, and Jiming Liu. 2015. Elderly Mobility and Daily Routine Analysis Based on Behavior-Aware Flow Graph Modeling. In *2015 International Conference on Healthcare Informatics*. 427–436. doi:10.1109/ICHI.2015.60

[119] Chen Li, William K Cheung, Jiming Liu, and Joseph K Ng. 2018. Automatic extraction of behavioral patterns for elderly mobility and daily routine analysis. *ACM Transactions on Intelligent Systems and Technology (TIST)* 9, 5 (2018), 1–26.

[120] Qimeng Li, Raffaele Gravina, Weilin Zang, Ye Li, Hongyu Wang, and Giancarlo Fortino. 2023. A Contactless Sensing Method for Social Activity Recognition. In *IEEE EUROCON 2023 - 20th International Conference on Smart Technologies*. 455–460. doi:10.1109/EUROCON56442.2023.10199041

[121] Shengjie Li, Zhaopeng Liu, Qin Lv, Yanyan Zou, Yue Zhang, and Daqing Zhang. 2025. WiLife: Long-Term Daily Status Monitoring and Habit Mining of the Elderly Leveraging Ubiquitous Wi-Fi Signals. *ACM Transactions on Computing for Healthcare* 6, 1 (2025), 1–29.

[122] Jonathan Liono, Zahraa S Abdallah, A Kai Qin, and Flora D Salim. 2018. Inferring transportation mode and human activity from mobile sensing in daily life. In *Proceedings of the 15th EAI International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services*. 342–351.

[123] Jonathan Liono, Mohammad Saiedur Rahaman, Flora D Salim, Yongli Ren, Damiano Spina, Falk Scholer, Johanne R Trippas, Mark Sanderson, Paul N Bennett, and Ryen W White. 2020. Intelligent task recognition: Towards enabling productivity assistance in daily life. In *Proceedings of the 2020 International Conference on Multimedia Retrieval*. 472–478.

[124] Haobing Liu, Yanmin Zhu, Tianzi Zang, Jiadi Yu, and Haibin Cai. 2020. Jointly modeling individual student behaviors and social influence for prediction tasks. In *proceedings of the 29th ACM international conference on Information & Knowledge Management*. 865–874.

[125] Peng Liu, Tun Lu, and Ning Gu. 2021. Behavior Analysis and Prediction of Disabled in Smart Home. In *2021 IEEE 24th International Conference on Computer Supported Cooperative Work in Design (CSCWD)*. 1027–1032. doi:10.1109/CSCWD49262.2021.9437754

[126] Jin Lu, Chao Shang, Chaoqun Yue, Reynaldo Morillo, Shweta Ware, Jayesh Kamath, Athanasios Bamis, Alexander Russell, Bing Wang, and Jinbo Bi. 2018. Joint modeling of heterogeneous sensing data for depression assessment via multi-task learning. *Proceedings of the ACM on interactive, mobile, wearable and ubiquitous technologies* 2, 1 (2018), 1–21.

[127] Jianchao Lu, Xi Zheng, Michael Sheng, Jiong Jin, and Shui Yu. 2020. Efficient Human Activity Recognition Using a Single Wearable Sensor. *IEEE Internet of Things Journal* 7, 11 (2020), 11137–11146. doi:10.1109/JIOT.2020.2995940

[128] Cong Ma, Huy Quan Vu, Jinlong Wang, Van-Hau Trieu, and Gang Li. 2024. Understanding Residents' Behavior for Smart City Management by Sequential and Periodic Pattern Mining. *IEEE Transactions on Computational Social Systems* 11, 1 (2024), 1260–1276. doi:10.1109/TCSS.2023.3249740

[129] Zixiang Ma, John Bigham, Stefan Poslad, Bang Wu, Xiaoshuai Zhang, and Eliane Bodanese. 2018. Device-Free, Activity During Daily Life, Recognition Using a Low-Cost Lidar. In *2018 IEEE Global Communications Conference (GLOBECOM)*. 1–6. doi:10.1109/GLOCOM.2018.8647251

[130] Rabin Maharjan, Koichi Shiraishi, Takehiro Yamamoto, Yusuke Yamamoto, and Hiroaki Ohshima. 2019. Development of IoT monitoring device and prediction of daily life behavior. In *Proceedings of the 21st International Conference on Information Integration and Web-based Applications & Services*. 584–588.

[131] Alina Matei, Andreea Glavan, Petia Radeva, and Estefania Talavera. 2021. Towards Eating Habits Discovery in Egocentric Photo-Streams. *IEEE Access* 9 (2021), 17495–17506. doi:10.1109/ACCESS.2021.3053175

[132] Mark A. McDaniel and Gilles O. Einstein. 2000. Strategic and automatic processes in prospective memory retrieval: a multiprocess framework. *Applied Cognitive Psychology* 14, 7 (2000), S127–S144. arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1002/acp.775 doi:10.1002/acp.775

[133] Sakorn Mekruksavanich, Ponnipa Jantawong, Wikanda Phaphan, and Anuchit Jitpattanakul. 2024. A Sensor-Based Deep Learning Approach for Recognizing Daily and Work Activities in Open Environments for Sanitation Workers. In *2024 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunications Engineering (ECTI DAMT & NCON)*. 577–581. doi:10.1109/ECTIDAMTNCON60518.2024.10480100

[134] Shinya Misaki, Keisuke Umakoshi, Tomokazu Matsui, Hyuckjin Choi, Manato Fujimoto, and Keiichi Yasumoto. 2021. Non-contact in-home activity recognition system utilizing Doppler sensors. In *Adjunct Proceedings of the 2021 International Conference on Distributed Computing and Networking*. 169–174.

[135] Prabhat Mishra, Shiva Murthy Busetty, and Suresh Kumar Gudla. 2020. Enhanced Activity Recognition of the IoT Smart Home users through Cluster Analysis. In *2020 IEEE 6th World Forum on Internet of Things (WF-IoT)*. 1–6. doi:10.1109/WF-IoT48130.2020.9221177

[136] Prabhat Mishra, Suresh Kumar Gudla, Amogha D ShanBhag, and Joy Bose. 2019. Enhanced Alternate Action Recommender System Using Recurrent Patterns and Fault Detection System for Smart Home Users. In *2019 IEEE International Conference on Big Data (Big Data)*. 651–656. doi:10.1109/BigData47090.2019.9006571

[137] Alexandros Mitsou, Evaggelos Spyrou, and Theodoros Giannakopoulos. 2021. Multimodal Workplace Monitoring for Human Activity Recognition. In *Proceedings of the 25th Pan-Hellenic Conference on Informatics*. 206–211.

[138] Prakhar Mohan, Bogyeong Lee, Theodora Chaspari, and Changbum R. Ahn. 2021. Assessment of Daily Routine Uniformity in a Smart Home Environment Using Hierarchical Clustering. *IEEE Journal of Biomedical and Health Informatics* 25, 8 (2021), 3197–3208. doi:10.1109/JBHI.2020.3048327

[139] Karl A. Morris, Tania Giovannetti, and Sarah M. Lehman. 2020. Towards The Use of Smart Home Sensor Networks to Generate Predictive Activity Models. In *2020 IEEE 33rd International Symposium on Computer-Based Medical Systems (CBMS)*. 403–408. doi:10.1109/CBMS49503.2020.00083

[140] Alessandra Moschetti, Laura Fiorini, Dario Esposito, Paolo Dario, and Filippo Cavallo. 2017. Daily activity recognition with inertial ring and bracelet: An unsupervised approach. In *2017 IEEE International Conference on Robotics and Automation (ICRA)*. 3250–3255. doi:10.1109/ICRA.2017.7989370

[141] Soumaya Msaad, Yannick Zoetgnande, Joaquim Prud'homm, Geoffroy Cormier, and Guy Carrault. 2020. Frailty detection of older adults by monitoring their daily routine. In *2020 IEEE 20th International Conference on Bioinformatics and Bioengineering (BIBE)*. 701–704. doi:10.1109/BIBE50027.2020.00118

[142] Usman Naeem, Abdel-Rahman H Tawil, Ivans Semelis, Gaby Judah, and Robert Aunger. 2014. Inference of Hygiene Behaviours While Recognising Activities of Daily Living. In *Proceedings of the 3rd International Conference on Context-Aware Systems and Applications*. 154–161.

[143] Amnah Nasim, Micaela Morettini, Ilaria Marcantoni, Agnese Sbrollini, and Laura Burattini. 2019. Recurrence Analysis of Human Body Movements during Activities of Daily Living. In *2019 IEEE 23rd International Symposium on Consumer Technologies (ISCT)*. IEEE, 157–160.

[144] Shamila Nasreen, Muhammad Awais Azam, Usman Naeem, and Mustansar Ali Ghazanfar. 2015. Inference of activities with unexpected actions using pattern mining. In *Adjunct Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2015 ACM International Symposium on Wearable Computers*. 1479–1488.

[145] David Neal. 2007. A New Look at Habits and the Habit–Goal Interface. *Psychological Review* 114 (10 2007), 843–863. doi:10.1037/0033-295X.114.4.843

[146] Le T Nguyen, Patrick Tague, Ming Zeng, and Joy Zhang. 2015. Superad: Supervised activity discovery. In *Adjunct Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2015 ACM International Symposium on Wearable Computers*. 1463–1472.

[147] Thuong Nguyen, Qing Zhang, Duc V Le, and Mohan Karunanithi. 2017. Dirichlet process Gaussian mixture model for activity discovery in smart homes with ambient sensors. In *Proceedings of the 14th EAI International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services*. 383–392.

[148] Long Niu, Sachio Saiki, and Masahide Nakamura. 2018. A Preliminary Study for Qantitative Assessment of Life Rhythm Based on Sleeping and Eating Log Data. In *Proceedings of the 20th International Conference on Information Integration and Web-based Applications & Services*. 325–332.

[149] Shun Nomura, Yihsin Ho, and Toru Yamaguchi. 2013. Acquiring Personal Keyword Based on Communication Log for Pass-By Wireless Communication on Smartphone. In *2013 International Conference on Biometrics and Kansei Engineering*. 19–24. doi:10.1109/ICBAKE.2013.5

[150] Ehimwenma Nosakhare and Rosalind Picard. 2020. Toward assessing and recommending combinations of behaviors for improving health and well-being. *ACM Transactions on Computing for Healthcare* 1, 1 (2020), 1–29.

[151] Collins Poku Obeng, Benjamin Hu, Christopher Crocker, Ryan Striker, and Enrique Alvarez Vazquez. 2025. Handwashing and Showering Recognition Utilizing Wearable Humidity Sensor. In *2025 IEEE International Conference on Electro Information Technology (eIT)*. IEEE, 1–4.

[152] Harri Oinas-Kukkonen. 2010. Behavior Change Support Systems: A Research Model and Agenda. In *Persuasive Technology*, Thomas Ploug, Per Hasle, and Harri Oinas-Kukkonen (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 4–14.

[153] Matthew J Page, Joanne E McKenzie, Patrick M Bossuyt, Isabelle Boutron, Tammy C Hoffmann, Cynthia D Mulrow, Larissa Shamseer, Jennifer M Tetzlaff, Elie A Akl, Sue E Brennan, et al. 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *bmj* 372 (2021).

[154] Gaurav Paruthi, Shriti Raj, Natalie Colabianchi, Predrag Klasnja, and Mark W Newman. 2018. Finding the sweet spot (S) understanding context to support physical activity plans. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 2, 1 (2018), 1–17.

[155] Parvaneh Parvin, Stefano Chessa, Marco Manca, and Fabio Paterno'. 2018. Real-time anomaly detection in elderly behavior with the support of task models. *Proceedings of the ACM on human-computer interaction* 2, EICS (2018), 1–18.

[156] Hossein Pazhoumand-Dar, Martin Masek, and Chiou Peng Lam. 2015. Unsupervised monitoring of electrical devices for detecting deviations in daily routines. In *2015 10th International Conference on Information, Communications and Signal Processing (ICICS)*. 1–6. doi:10.1109/ICICS.2015.7459849

[157] Brian Pentland and Martha Feldman. 2005. Organizational Routines as a Unit of Analysis. *Industrial and Corporate Change* 14 (02 2005), 793–815. doi:10.1093/icc/dth070

[158] Sergio Lluva Plaza, José M. Villadangos Carrizo, Ana Jiménez Martín, Antonio R. Jiménez Ruiz, and Juan Jesús García Domínguez. 2022. Routine analysis of community-dwelling older adults and anomaly detection by using room level location. In *2022 IEEE International Symposium on Medical Measurements and Applications (MeMeA)*. 1–2. doi:10.1109/MeMeA54994.2022.9856559

[159] Sergio Lluva Plaza, Raúl Montoliu Colás, Ana Jiménez Martín, José M. Villadangos Carrizo, Emilio Sansano-Sansano, and Juan Jesús García Domínguez. 2022. BLE RSSI Database for Analyzing Routines of Community-Dwelling Older Adults. In *2022 14th International Congress on Ultra Modern Telecommunications and Control Systems and Workshops (ICUMT)*. 50–55. doi:10.1109/ICUMT57764.2022.9943410

[160] Lorenzo Porzi, Stefano Messelodi, Carla Mara Modena, and Elisa Ricci. 2013. A smart watch-based gesture recognition system for assisting people with visual impairments. In *Proceedings of the 3rd ACM International Workshop on Interactive Multimedia on Mobile & Portable Devices* (Barcelona, Spain) *(IMMPD '13)*. Association for Computing Machinery, New York, NY, USA, 19–24. doi:10.1145/2505483.2505487

[161] Ada Pozo, Thanh-Trung Phan, and Daniel Gatica-Perez. 2020. Learning urban nightlife routines from mobile data. In *Proceedings of the 19th International Conference on Mobile and Ubiquitous Multimedia*. 107–118.

[162] Tian Qin, Wufan Shangguan, Guojie Song, and Jie Tang. 2018. Spatio-temporal routine mining on mobile phone data. *ACM Transactions on Knowledge Discovery from Data (TKDD)* 12, 5 (2018), 1–24.

[163] Shigang Quan, Shui Liu, Zhenzhe Zheng, and Fan Wu. 2023. Enhancing repeat-aware recommendation from a temporal-sequential perspective. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*. 2095–2105.

[164] Zeinab R. Yousefi, Tung Vuong, Marie Alghossein, Tuukka Ruotsalo, Giulio Jaccuci, and Samuel Kaski. 2024. Entity Footprinting: Modeling Contextual User States via Digital Activity Monitoring. *ACM Transactions on Interactive Intelligent Systems* 14, 2 (2024), 1–27.

[165] Ghazala Rafiq, Rafiq Muhammad, and Soon Ju Kang. 2024. Self-Attention Enhanced Spatio-Temporal Convolution for ADL Recognition in Elderly Care. In *2024 15th International Conference on Information and Communication Technology Convergence (ICTC)*. 1520–1525. doi:10.1109/ICTC62082.2024.10827770

[166] Parisa Rashidi, Diane J. Cook, Lawrence B. Holder, and Maureen Schmitter-Edgecombe. 2011. Discovering Activities to Recognize and Track in a Smart Environment. *IEEE Transactions on Knowledge and Data Engineering* 23, 4 (2011), 527–539. doi:10.1109/TKDE.2010.148

[167] Yordan P. Raykov, Luc J. W. Evers, Reham Badawy, Bastiaan R. Bloem, Tom M. Heskes, Marjan J. Meinders, Kasper Claes, and Max A. Little. 2021. Probabilistic Modelling of Gait for Robust Passive Monitoring in Daily Life. *IEEE Journal of Biomedical and Health Informatics* 25, 6 (2021), 2293–2304. doi:10.1109/JBHI.2020.3037857

[168] Andreas Reckwitz. 2002. Toward a Theory of Social Practices: A Development in Culturalist Theorizing. *European Journal of Social Theory* 5, 2 (2002), 243–263. arXiv:https://doi.org/10.1177/13684310222225432 doi:10.1177/13684310222225432

[169] Attila Reiss and Didier Stricker. 2011. Towards global aerobic activity monitoring. In *Proceedings of the 4th International Conference on PErvasive Technologies Related to Assistive Environments* (Heraklion, Crete, Greece) *(PETRA '11)*. Association for Computing Machinery, New York, NY, USA, Article 12, 8 pages. doi:10.1145/2141622.2141637

[170] Attila Reiss, Didier Stricker, and Gustaf Hendeby. 2013. Towards robust activity recognition for everyday life: Methods and evaluation. In *2013 7th International Conference on Pervasive Computing Technologies for Healthcare and Workshops*.

25–32. doi:10.4108/icst.pervasivehealth.2013.251928

[171] Daniele Riboni, Claudio Bettini, Gabriele Civitarese, Zaffar Haider Janjua, and Viola Bulgari. 2015. From lab to life: Fine-grained behavior monitoring in the elderly's home. In *2015 IEEE International Conference on Pervasive Computing and Communication Workshops (PerCom Workshops)*. 342–347. doi:10.1109/PERCOMW.2015.7134060

[172] Mariana Jacob Rodrigues, Octavian Postolache, and Francisco Cercas. 2025. Air Quality Monitoring for Human Activity Recognition. In *2025 IEEE Medical Measurements & Applications (MeMeA)*. 1–6. doi:10.1109/MeMeA65319.2025.11067996

[173] Aaqib Saeed, Tanir Ozcelebi, Stojan Trajanovski, and Johan J. Lukkien. 2019. End-to-End Multi-Modal Behavioral Context Recognition in a Real-Life Setting. In *2019 22th International Conference on Information Fusion (FUSION)*. 1–8. doi:10.23919/FUSION43075.2019.9011194

[174] Yuiko Sakuma, Sofia Kleisarchaki, Levent Gürgen, and Hiroaki Nishi. 2019. Exploring Variability in IoT Data for Human Activity Recognition. In *IECON 2019 - 45th Annual Conference of the IEEE Industrial Electronics Society*, Vol. 1. 5312–5318. doi:10.1109/IECON.2019.8927472

[175] Dhanish Semar, Ted Chen, and Rashina Hoda. 2015. Automating SmartPhone Automation: Enhancing User Experience with Behaviour Learning. In *2015 24th Australasian Software Engineering Conference*. 108–117. doi:10.1109/ASWEC.2015.22

[176] Hela Sfar and Amel Bouzeghoub. 2019. DataSeg: Dynamic streaming sensor data segmentation for activity recognition. In *Proceedings of the 34th ACM/SIGAPP Symposium on Applied Computing*. 557–563.

[177] Zahraa Khais Shahid, Saguna Saguna, and Christer Åhlund. 2023. Forecasting and Detecting Anomalies in ADLs in Single-Resident Elderly Smart Homes. In *Proceedings of the 2023 International Conference on Research in Adaptive and Convergent Systems*. 1–8.

[178] Meng Shang, Lenore Dedeyne, Jolan Dupont, Laura Vercauteren, Nadjia Amini, Laurence Lapauw, Evelien Gielen, Sabine Verschueren, Carolina Varon, Walter De Raedt, and Bart Vanrumste. 2024. Otago Exercises Monitoring for Older Adults by a Single IMU and Hierarchical Machine Learning Models. *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 32 (2024), 462–471. doi:10.1109/TNSRE.2024.3355299

[179] Manan Sharma, Shivam Tiwari, Suchetana Chakraborty, and Dip Sankar Banerjee. 2020. Behavior Analysis through Routine Cluster Discovery in Ubiquitous Sensor Data. In *2020 International Conference on COMmunication Systems & NETworkS (COMSNETS)*. 267–274. doi:10.1109/COMSNETS48256.2020.9027377

[180] Elizabeth Shove, Matt Watson, and Mika Pantzar. 2012. The dynamics of social practice: Everyday life and how it changes. (2012).

[181] Prakhar Shukla, Parnab Kumar Chanda, Ramnath Jayachandran, and Ashok Subash. 2018. A Framework for User Routine Discovery in Smart Homes. In *2018 IEEE 6th International Conference on Future Internet of Things and Cloud (FiCloud)*. 288–295. doi:10.1109/FiCloud.2018.00049

[182] B. F. Skinner. 1953. *Science and Human Behavior*. Macmillan, New York, NY.

[183] Pouneh Soleimaninejadian, Min Zhang, Yiqun Liu, and Shaoping Ma. 2018. Mood Detection and Prediction Based on User Daily Activities. In *2018 First Asian Conference on Affective Computing and Intelligent Interaction (ACII Asia)*. 1–6. doi:10.1109/ACIIAsia.2018.8470365

[184] Lucy A. Suchman. 1987. *Plans and situated actions: the problem of human-machine communication*. Cambridge University Press, USA.

[185] Yu Sun, Nicholas Jing Yuan, Xing Xie, Kieran McDonald, and Rui Zhang. 2017. Collaborative intent prediction with real-time contextual data. *ACM Transactions on Information Systems (TOIS)* 35, 4 (2017), 1–33.

[186] Vedant Das Swain, Manikanta D Reddy, Kari Nies, Louis Tay, Munmun De Choudhury, and Gregory D Abowd. 2019. Birds of a Feather Clock Together: A Study of Person-Organization Fit Through Latent Activity Routines. *Proc. ACM Hum. Comput. Interact.* 3, CSCW (2019), 165–1.

[187] John Sweller. 1988. Cognitive load during problem solving: Effects on learning. *Cognitive Science* 12, 2 (1988), 257–285. doi:10.1016/0364-0213(88)90023-7

[188] Wataru Takano. 2020. Annotation Generation From IMU-Based Human Whole-Body Motions in Daily Life Behavior. *IEEE Transactions on Human-Machine Systems* 50, 1 (2020), 13–21. doi:10.1109/THMS.2019.2960630

[189] Wataru Takano. 2020. Sentence Generation from IMU-based Human Whole-Body Motions in Daily Life Behaviors. In *2020 IEEE/SICE International Symposium on System Integration (SII)*. IEEE, 669–674.

[190] Dalai Tang, Yuri Yoshihara, Takenori Obo, Takahiro Takeda, Janos Botzheim, and Naoyuki Kubota. 2016. Evolution strategy for anomaly detection in daily life monitoring of elderly people. In *2016 55th Annual Conference of the Society of Instrument and Control Engineers of Japan (SICE)*. 1376–1381. doi:10.1109/SICE.2016.7749272

[191] Romina Torres, Mauricio Poblete, and Rodrigo Salas. 2017. Classifying human actions in daily life using computational intelligence techniques. In *2017 CHILEAN Conference on Electrical, Electronics Engineering, Information and Communication Technologies (CHILECON)*. 1–5. doi:10.1109/CHILECON.2017.8229514

[192] Quang-Linh Tran, Binh Nguyen, Gareth JF Jones, and Cathal Gurrin. 2024. MemoriLens: a Low-cost Lifelog Camera Using Raspberry Pi Zero. In *Proceedings of the 2024 International Conference on Multimedia Retrieval*. 1255–1259.

[193] Cheng-Hung Tsai, Cheng-Hao Chu, Sun-Wei Liu, Sun-Yuan Hsieh, and Vincent S. Tseng. 2017. Mining Life Patterns from Wearable Sensors Data for Elderly Anomaly Detection. In *2017 Conference on Technologies and Applications of Artificial Intelligence (TAAI)*. 66–71. doi:10.1109/TAAI.2017.43

[194] Md Zia Uddin. 2014. A robust daily human activity recognition and prediction system. In *Proceedings of the 2014 International C* Conference on Computer Science & Software Engineering*. 1–8.

[195] Dhruv Deepak Upadhyay, Vaibhav Pandey, Nitish Nag, and Ramesh Jain. 2020. Personalized user modelling for sleep insight. In *Proceedings of the 1st International Workshop on Human-centric Multimedia Analysis*. 13–20.

[196] Maitri Vaghela and Kalyan Sasidhar. 2020. Analyzing the human behavior using pervasive sensing system. In *Proceedings of the 21st International Conference on Distributed Computing and Networking*. 1–5.

[197] Yonatan Vaizman, Katherine Ellis, and Gert Lanckriet. 2017. Recognizing Detailed Human Context in the Wild from Smartphones and Smartwatches. *IEEE Pervasive Computing* 16, 4 (2017), 62–74. doi:10.1109/MPRV.2017.3971131

[198] Nikhil Verma, Sona Mundody, and Ram Mohana Reddy Guddeti. 2024. An Efficient AI and IoT Enabled System for Human Activity Monitoring and Fall Detection. In *2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)*. IEEE, 1–6.

[199] Bo Wang, Xie Ni, Guoru Zhao, Yingnan Ma, Xing Gao, Huiqi Li, Cuiju Xiong, Lei Wang, and Shengyun Liang. 2017. A wearable action recognition system based on acceleration and attitude angles using real-time detection algorithm. In *2017 39th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE, 2385–2389.

[200] Benyou Wang, Sihai Zhang, Xia Peng, and Zhongyi Zhang. 2021. Campus Behavior Analyses of College Students Based on Large-scale Data. In *2021 IEEE International Conference on Power Electronics, Computer Applications (ICPECA)*. 349–354. doi:10.1109/ICPECA51329.2021.9362568

[201] Guoquan Wang, Qiang Luo, Weisong Hu, Pengfei Yao, Wencong Zeng, Guorui Zhou, and Kun Gai. 2025. FIM: Frequency-Aware Multi-View Interest Modeling for Local-Life Service Recommendation. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 1748–1757.

[202] Xue Wang and Yang Zhang. 2024. TextureSight: Texture Detection for Routine Activity Awareness with Wearable Laser Speckle Imaging. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 4 (2024), 1–27.

[203] Yan Wang, Shuang Cang, and Hongnian Yu. 2018. A Data Fusion-Based Hybrid Sensory System for Older People's Daily Activity and Daily Routine Recognition. *IEEE Sensors Journal* 18, 16 (2018), 6874–6888. doi:10.1109/JSEN.2018.2833745

[204] Mark Weiser. 1999. The computer for the 21st century. *SIGMOBILE Mob. Comput. Commun. Rev.* 3, 3 (July 1999), 3–11. doi:10.1145/329124.329126

[205] Bang Wu, Zixiang Ma, Stefan Poslad, and Yidong Li. 2018. WiFi Fingerprint Based, Indoor, Location-Driven Activities of Daily Living Recognition. In *2018 5th International Conference on Behavioral, Economic, and Socio-Cultural Computing (BESC)*. 148–151. doi:10.1109/BESC.2018.8697290

[206] Chia-Hua Wu and Ren-Hung Hwang. 2014. Routine Based User Classification. In *2014 IEEE International Conference on Internet of Things (iThings), and IEEE Green Computing and Communications (GreenCom) and IEEE Cyber, Physical and Social Computing (CPSCom)*. 444–449. doi:10.1109/iThings.2014.79

[207] Wenlong Wu, James M Keller, Marjorie Skubic, Mihail Popescu, and Kari R Lane. 2021. Early detection of health changes in the elderly using in-home multi-sensor data streams. *ACM Transactions on Computing for Healthcare* 2, 3 (2021), 1–23.

[208] Zongxing Xie, Bing Zhou, Xi Cheng, Elinor Schoenfeld, and Fan Ye. 2022. Passive and context-aware in-home vital signs monitoring using co-located uwb-depth sensor fusion. *ACM transactions on computing for healthcare* 3, 4 (2022), 1–31.

[209] Xuhai Xu, Xin Liu, Han Zhang, Weichen Wang, Subigya Nepal, Yasaman Sefidgar, Woosuk Seo, Kevin S Kuehn, Jeremy F Huckins, Margaret E Morris, et al. 2023. Globem: Cross-dataset generalization of longitudinal human behavior modeling. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 4 (2023), 1–34.

[210] Salisu Wada Yahaya, Ahmad Lotfi, and Mufti Mahmud. 2020. Towards the Development of an Adaptive System for Detecting Anomaly in Human Activities. In *2020 IEEE Symposium Series on Computational Intelligence (SSCI)*. 534–541. doi:10.1109/SSCI47803.2020.9308415

[211] Naoya Yamamoto, Takato Matsumoto, Tamami Sudo, Megumi Miyashita, and Toshiyuki Kondo. 2022. Ring-shaped wearable device for logging finger usage in daily life. In *2022 International Symposium on Micro-NanoMehatronics and Human Science (MHS)*. IEEE, 1–6.

[212] Jingkang Yang, Shuai Liu, Hongming Guo, Yuhao Dong, Xiamengwei Zhang, Sicheng Zhang, Pengyun Wang, Zitang Zhou, Binzhu Xie, Ziyue Wang, et al. 2025. Egolife: Towards egocentric life assistant. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 28885–28900.

[213] Chih-Ta Yen, Jia-Xian Liao, and Yi-Kai Huang. 2020. Human Daily Activity Recognition Performed Using Wearable Inertial Sensors Combined With Deep Learning Algorithms. *IEEE Access* 8 (2020), 174105–174114. doi:10.1109/ACCESS.2020.3025938

[214] Jie Yin, Qing Zhang, and Mohan Karunanithi. 2015. Unsupervised daily routine and activity discovery in smart homes. In *2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. 5497–5500. doi:10.1109/EMBC.2015.7319636

[215] Yuan Yuan, Jingtao Ding, Huandong Wang, and Depeng Jin. 2024. Generating daily activities with need dynamics. *ACM Transactions on Intelligent Systems and Technology* 15, 2 (2024), 1–28.

[216] Songhui Yue, Randy Smith, and Songqing Yue. 2017. A state-based approach to context modeling and computing. In *2017 IEEE SmartWorld, Ubiquitous Intelligence & Computing, Advanced & Trusted Computed, Scalable Computing & Communications, Cloud & Big Data Computing, Internet of People and Smart City Innovation (SmartWorld/SCALCOM/UIC/ATC/CBDCom/IOP/SCI)*. 1–6. doi:10.1109/UIC-ATC.2017.8397532

[217] Onur Yürüten, Jiyong Zhang, and Pearl HZ Pu. 2014. Predictors of life satisfaction based on daily activities from mobile sensor data. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 497–500.

[218] Marsil Zakour, Alaeddine Mellouli, and Rahul Chaudhari. 2021. HOIsim: Synthesizing Realistic 3D Human-Object Interaction Data for Human Activity Recognition. In *2021 30th IEEE International Conference on Robot & Human Interactive Communication (RO-MAN)*. 1124–1131. doi:10.1109/RO-MAN50785.2021.9515349

[219] Dorsaf Zekri, Ahmed Snoun, Thierry Delot, and Marie Thilliez. 2023. Context Sensitive Detection of Long-term Elderly Behavior Change. In *2023 International Conference on Innovations in Intelligent Systems and Applications (INISTA)*. 1–6. doi:10.1109/INISTA59065.2023.10310468

[220] Hengyang Zhang, Tao Huang, Yunjie Liu, Shixiang Zhu, Guan Gui, and Yuanying Chi. 2017. Senz: A context awareness middleware system used in mobile devices. In *2017 IEEE 85th Vehicular Technology Conference (VTC Spring)*. IEEE, 1–7.

[221] Meiyan Zhang, Dan Liu, Qisong Wang, Boqi Zhao, Ou Bai, and Jinwei Sun. 2022. Gait Pattern Recognition Based on Plantar Pressure Signals and Acceleration Signals. *IEEE Transactions on Instrumentation and Measurement* 71 (2022), 1–15. doi:10.1109/TIM.2022.3204088

[222] Yanxia Zhang, Andreas Girgensohn, and Yulius Tjahjadi. 2019. Activity Forecasting in Routine Tasks by Combining Local Motion Trajectories and High-Level Temporal Models. In *2019 IEEE International Conference on Computational Science and Engineering (CSE) and IEEE International Conference on Embedded and Ubiquitous Computing (EUC)*. 447–452. doi:10.1109/CSE/EUC.2019.00089

[223] Yiyuan Zhang, Jorgen Wullems, Ine D'Haeseleer, Vero Vanden Abeele, and Bart Vanrumste. 2020. Bathroom activity monitoring for older adults via wearable device. In *2020 IEEE International Conference on Healthcare Informatics (ICHI)*. 1–10. doi:10.1109/ICHI48887.2020.9374396

[224] Jiangchuan Zheng, Siyuan Liu, and Lionel M. Ni. 2014. Effective Mobile Context Pattern Discovery via Adapted Hierarchical Dirichlet Processes. In *2014 IEEE 15th International Conference on Mobile Data Management*, Vol. 1. 146–155. doi:10.1109/MDM.2014.24

[225] Shuang Zhou, Lingchao Guo, Zhaoming Lu, Xiangming Wen, and Zijun Han. 2023. Wi-Monitor: Daily Activity Monitoring Using Commodity Wi-Fi. *IEEE Internet of Things Journal* 10, 2 (2023), 1588–1604. doi:10.1109/JIOT.2022.3210378

[226] Chun Zhu and Weihua Sheng. 2010. Recognizing human daily activity using a single inertial sensor. In *2010 8th World Congress on Intelligent Control and Automation*. 282–287. doi:10.1109/WCICA.2010.5555072